

National Transportation Safety Board

Office of Aviation Safety

Washington, DC 20594



DCA25MA108

AIR TRAFFIC CONTROL / HUMAN PERFORMANCE

Group Chairman's Factual Report

June 18, 2025

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A. ACCIDENT

Location: Washington, District of Columbia
Date: January 29, 2025
Time: 2048 eastern standard time (EST)¹
0148 coordinated universal time (UTC)² / January 30, 2025
Airplane: Mitsubishi Heavy Industries (MHI) RJ Aviation CL-600-2C10, N709PS,
PSA Airlines flight 5342 (JIA5342)
Helicopter: Sikorsky UH-60L, Tail No. 00-26860, US Army, Priority Air Transport
flight 25 (PAT25)

B. AIR TRAFFIC CONTROL AND HUMAN PERFORMANCE GROUP

Group Chairman	Brian Soper National Transportation Safety Board Washington, DC
Group Co-Chairman - ATC	Sarah Lewis National Transportation Safety Board Washington, DC
Group Co-Chairman - HP	Katherine Wilson National Transportation Safety Board Washington, DC
Group Member	James Clothier Federal Aviation Administration Westbury, NY
Group Member	Dan Carrico National Air Traffic Controllers Association Chicago, IL

C. SUMMARY

On January 29, 2025, about 2048 eastern standard time (EST), a Sikorsky UH-60L, operated by the US Army under the callsign PAT25, and an MHI (Mitsubishi Heavy Industries) RJ Aviation (formerly Bombardier) CL-600-2C10 (CRJ700), N709PS, operated by PSA Airlines as flight 5342, collided in flight approximately 0.5 miles

¹ All times are in eastern standard time (EST) unless otherwise noted.

² UTC - coordinated universal time is the international time standard (formerly Greenwich Mean Time, or GMT). Zero hours UTC is midnight in Greenwich, England, which is located at 0 degrees longitude. Everything east of Greenwich (up to 180 degrees) is later in time; everything west is earlier. For example, the EST zone in the U.S. is UTC minus (-) 5. Therefore, the accident time in Washington, DC was 2048 EST, making the UTC time 0148 UTC the following day.

southeast of Ronald Reagan Washington National Airport (DCA), Arlington, Virginia, and impacted the Potomac River in southwest Washington, District of Columbia.

The 2 pilots, 2 flight attendants, and 60 passengers aboard the airplane and all 3 crew members aboard the helicopter were fatally injured. Both aircraft were destroyed as a result of the accident. Flight 5342 was operating under the provisions of Title 14 *Code of Federal Regulations (CFR)* Part 121 as a scheduled domestic passenger flight from Wichita Dwight D. Eisenhower National Airport (ICT), Wichita, Kansas, to DCA. PAT25 originated from Davison Army Airfield (DAA), Fort Belvoir, Virginia, for the purpose of the pilot's annual standardization evaluation with the use of night vision goggles (NVGs). Night visual meteorological conditions prevailed in the area of DCA at the time of the accident.

D. DETAILS OF THE INVESTIGATION

The air traffic control (ATC) and human performance (HP) workgroup formed on Wednesday, January 29, 2025, and consisted of the group chairman and co-chairman from AS-30 ATC, co-chairman from AS-60 Human Performance (HP), and group members from the National Air Traffic Controllers Association (NATCA) and Federal Aviation Administration (FAA).

On January 30, 2025, the group traveled to Arlington, VA, and met at the command post set up in hangar 6 at DCA. The group traveled to the DCA airport traffic control tower (ATCT) and was provided an event briefing by the air traffic manager (ATM) and members of their staff. Also present were members from the FAA's safety intelligence and response group (SIRG), eastern service area quality control group (ESA QCG), and attorneys from the FAA's office of chief counsel. The group then attended the organizational meeting at the command post and concluded group activities for the day.

On January 31, 2025, the group attended the morning planning meeting at the command post, then traveled to FAA headquarters (HQ) and conducted interviews with the DCA ATCT local control (LC) and ground control (GC) controllers that were working in the tower at the time of the accident. The group returned to the command post and attended the evening progress meeting before concluding group activities for the day.

On February 1, 2025, the group attended the morning planning meeting at the command post, then traveled to Dulles International Airport (IAD) and conducted an interview with the assistant local control (ALC) controller that was working in the tower at the time of the accident. The group then returned to the command post and attended the evening progress meeting, then returned to DCA ATCT to photograph pertinent tower equipment, workstations and viewpoints from the tower cab under

night conditions similar to those present at the time of the accident and concluded group activities for the day.

On February 2, 2025, the group attended the morning planning meeting at the command post, then returned to DCA ATCT and conducted an interview with the operations supervisor (OS) that was working in the tower at the time of the accident. The group then collected and reviewed additional data then returned to the command post and attended the evening progress meeting before concluding group activities for the day.

On February 3, 2025, the group attended the morning planning meeting at the command post, then returned to DCA ATCT and conducted interviews with the clearance delivery (CD) controller that was working at the time of the accident, and the operations manager (OM) who was not present in the facility at the time of the accident. The group then returned to the command post and attended the evening progress meeting. After the progress meeting, the group chairman and co-chairmen met at NTSB HQ and auditioned the cockpit voice recorder (CVR) from JIA5342 and concluded group activities for the day.

On February 4, 2025, the group attended the morning planning meeting at the command post, then returned to DCA ATCT and conducted an interview with the DCA ATCT ATM who was not present in the facility at the time of the accident. The group then returned to the command post, worked on field notes, attended the evening progress meeting, and concluded group activities for the day.

On February 5, 2025, the group met at DCA ATCT and conducted an interview with the Potomac Terminal Radar Approach Control (PCT TRACON) ATM then the group chairman and co-chairmen met at NTSB HQ and auditioned the CVR from PAT25. The group then returned to the command post and attended the evening progress meeting before concluding group activities for the day.

On February 6, 2025, the group met at the command post, completed field notes and obtained concurrence and provided them to the investigator in-charge (IIC) and deputy IIC for review.

On February 7, 2025, the group chairmen met at the command post for the morning planning meeting and then met with personnel from the FAA's office of aeronautical information services (AJV-A2) to discuss information surrounding the Baltimore-Washington Helicopter Route Chart. The field notes were concurred upon by the group, approved by the IIC and the group was released.

On February 8-10, 2025, the group chairman and co-chairmen remained on-site and continued follow-up work and assistance to the other groups and parties to the investigation, before being released and returning home on February 11, 2025.

On the evening of March 26, the group chairmen met at DCA ATCT to participate in a visibility study conducted by the NTSB Research and Engineering division. The group members coordinated and oversaw video documentation of the helicopter flight operations from the tower perspective. The visibility study is still in work and details will be provided in a separate report later.

On March 27, 2025, the entire group and the helicopter operations (OPS) and HP group chairman traveled to the DCA ATCT to interview two ATC personnel who were involved with the DCA helicopter work group (HWG). Prior to the first interview, the facility had a non-safety related emergency. As a result, one interview was postponed, and the other interview was moved and conducted at NTSB HQ. The group worked on interview notes and concluded group activities for the day. That evening the group chairmen met at DCA ATCT to participate in the second night of the visibility study work. The group coordinated and oversaw video documentation of the CRJ flight operations from the tower perspective.

On March 28, 2025, the group traveled to NTSB HQ and conducted interviews with the Washington district traffic management officer (TMO) and the PCT TRACON, Mount Vernon sector OM. The group completed field notes, obtained concurrence, and the group was released

On April 14-15, 2025, the group traveled to the FAA William J. Hughes Technical Center (WJHTC) in Atlantic City, NJ. The group visited various laboratories, attended briefings by various subject matter experts. Details of this visit are described in section I of this report.

On April 22, 2025, the group conducted a supplemental follow-up interview with the DCA ATCT OM. This supplemental interview was the one previously postponed and focused on the DCA ATCT OM's involvement with the DCA HWG.

E. FACTUAL INFORMATION

1.0 History of Flight

PAT25 was provided ATC services from DAA ATCT, Quantico Radar Approach Control (NYG RAPCON), PCT TRACON, and DCA ATCT. JIA5342 was provided ATC services from ICT ATCT, Kansas City Air Route Traffic Control Center (ZKC ARTCC), Indianapolis (ZID) ARTCC, Washington (ZDC) ARTCC, PCT TRACON, and DCA ATCT. The history of flight provided below focuses primarily on the services provided by DCA ATCT and touches only briefly on the other ATC facilities for the purpose of timeline continuity. Preflight planning is covered in the respective Operations and Human Performance Group Chairman's factual reports in the docket.

The history of flight was constructed using surveillance data (see subsection 2.0 of this section), audio recordings³, and the aircraft accident package⁴ provided by the FAA. The [yet to be titled combined audio transcript]⁵ was also used, as well as information gleaned in post-accident interviews conducted with involved ATC personnel (see section G, sub-section 1.0).

About 1833, JIA5342 contacted ICT ATCT and requested taxi clearance and GC instructed them to taxi to runway 19R. About 1838, ICT ATCT cleared JIA5342 for takeoff from runway 19R and according to surveillance data the airplane became airborne about 1839.

About 1842, PAT25 contacted DAA ATCT and requested taxi instructions for a southeast departure from runway 32. About 1843, DAA ATCT cleared PAT25 for takeoff from runway 32 and according to surveillance data the helicopter became airborne about 1845.

About 1843, JIA5342 was handed off⁶ to, and contacted ZKC ARTCC still climbing to a cruise altitude of flight level (FL)⁷ 330 and transitioning through multiple sectors until being handed off to ZID ARTCC.

About 1846, DAA ATCT asked PAT25 if they would be contacting DCA ATCT, and there was no response from PAT25.

About 1848, DAA ATCT asked PAT25 if they would be contacting NYG RAPCON, and PAT25 advised they would be contacting PCT TRACON. DAA ATCT approved their frequency change to PCT TRACON.

About 1849, PAT25 attempted to contact PCT TRACON but they were still on DAA ATCT's frequency. DAA ATCT instructed PAT25 to check their frequency and to contact PCT TRACON on 118.95. There was no response from PAT25.

About 1859, PAT25 contacted DAA ATCT and stated, "pat two five we'll stay up with you if...", the transmission seemed to be cutoff mid transmission, however there were no further communications recorded between PAT25 and DAA ATCT.

About 1931, JIA5342 was handed off to, and contacted ZID ARTCC level at FL330 and transitioned through multiple sectors. During their transition,

³ Certified ATC audio recordings are in Attachment 1: ATC Audio Recordings.

⁴ The FAA Aircraft Accident Package is in Attachment 2: FAA Aircraft Accident Package.

⁵ The combined audio transcript has not yet been finalized but will be included in the Recorders Factual Report information in the docket.

⁶ A handoff is an action taken to transfer the radar identification of an aircraft from one controller to another if the aircraft will enter the receiving controller's airspace and radio communications with the aircraft will be transferred.

⁷ Flight Level (FL) – A level of constant atmospheric pressure related to a reference datum of 29.92 inches of mercury. Each is stated in three digits that represent hundreds of feet. For example, flight level (FL) 250 represents a barometric altimeter indication of 25,000 feet; FL 255, an indication of 25,500 feet.

JIA5342 received minor amendments to their clearance and a climb to FL370 before being handed off to ZDC ARTCC.

About 2007, JIA5342 was handed off to, and contacted ZDC ARTCC level at FL370 and transitioned through multiple sectors. During their transition, JIA5342 received minor amendments to the clearance until being cleared to descend via the TRUPS Five Arrival⁸ into DCA.

About 2018, JIA5342 was advised by ZDC ARTCC to expect moderate turbulence at 12,000 feet and below while on approach to DCA.

About 2021, JIA5342 was handed off to and contacted PCT TRACON descending on the TRUPS Five Arrival and advised they had ATIS⁹ information alpha (A). PCT TRACON advised them to expect light chop through 4,000 feet on their descent.

About 2032, PAT25 contacted the DCA ATCT LC controller and requested to transition via the Cabin John Bridge, route 1, route 4, to DAA¹⁰. The LC controller issued the DCA altimeter setting of 29.89 and requested their position. The crew of PAT25 read back the altimeter setting and the remainder of the transmission was indiscernible.

About 2033, the LC controller instructed PAT25 to ident¹¹ and subsequently advised them they were radar contact¹² six miles north of Bethesda, MD, at 1,200 feet and asked them to repeat their request. The crew of PAT25 restated their request to transition via the Cabin John Bridge, route 1, route 4, to DAA. The LC controller acknowledged and approved their request.

About 2039, after being provided with a routine set-up for the arrival into DCA, JIA5342 reported the Potomac River in sight. PCT TRACON cleared JIA5342 for the Mount Vernon Visual Runway 1 Approach¹³ and instructed them to

⁸ The published Standard Terminal Arrival (STAR) used by flight crews of aircraft arriving DCA and Andrews (ADW) is included in Attachment 3: DCA Approach and Arrival Procedures.

⁹ ATIS - Automatic Terminal Information Service - A continuous broadcast of current, routine information to arriving and departing aircraft by means of continuous and repetitive broadcasts throughout the day or specified portion of the day.

¹⁰ According to post-accident interviews, this was a routine VFR helicopter routing requested by crews transitioning through DCA class B airspace from north to south as in this case. Routes 1 and 4 referring VFR helicopter routes as published in the Baltimore-Washington Helicopter Route Chart at the time of the accident (dated December 26, 2024).

¹¹ Ident - A request for a pilot to activate the aircraft transponder identification feature. This will help the controller to confirm an aircraft identity or to identify an aircraft and is a common means of establishing radar contact by ATC.

¹² Radar Contact - A term used by ATC to inform an aircraft that it is identified using an approved ATC surveillance source on an air traffic controllers display and that radar flight following will be provided until radar service is terminated. Radar service may also be provided within the limits of necessity and capability. When a pilot is informed of "radar contact," he/she automatically discontinues reporting over compulsory reporting points

¹³ The published visual approach procedure for aircraft arriving DCA when the airport is in a north configuration is included in Attachment 3: DCA Approach and Arrival Procedures.

maintain 170 knots until five miles out. JIA5342 acknowledged the clearance and speed restriction.

About 2043, JIA5342 was handed off to, and contacted DCA ATCT inbound on the Mount Vernon Visual Runway 1 approach. The LC controller issued the winds of 320° at 17 knots with gusts to 25 knots and asked if they could accept runway 33.

About 2044, the crew of JIA5342 advised they could accept the runway change to runway 33 and the LC controller then advised them to change to, and circle to runway 33 at the Wilson Bridge, then cleared them to land on runway 33. The crew read back the runway change instructions and landing clearance to runway 33.

About 2045, the crew of PAT25 reported over the Memorial Bridge (a compulsory reporting point located along helicopter route 1) and the LC controller acknowledged the report.

About 2046, the LC controller advised the crew of PAT25 of traffic just south of the Wilson Bridge, a CRJ [JIA5342] at 1,200 feet circling to runway 33. The crew acknowledged the traffic advisory stating immediately that they had the traffic in sight and requested visual separation¹⁴. The LC controller approved their request for visual separation.

About 2047, the LC controller asked the crew of PAT25 if they still had the previously called traffic in sight and instructed them to pass behind the CRJ. The conflict alert (CA)¹⁵ audible alarm could be heard in the background when the LC was keyed up. The crew again reported the traffic in sight and requested visual separation. The LC controller approved their request for visual separation.

[According to radar data, the CA began at 2047:33 and remained until 2047:52 then began again at 2047:59 and remained active through the time of the accident.]

About 2048, while the LC controller was issuing a landing clearance to an uninvolved aircraft, the CA audible alarm continued to be heard in the background followed by audible reactions from other ATC personnel in the tower as the midair collision occurred.

¹⁴ Visual Separation - A means employed by ATC to separate aircraft in terminal areas and en route airspace in the NAS. In this case, by the pilot seeing the other aircraft involved and upon instructions from the controller providing their own separation by maneuvering their aircraft as necessary to avoid it. This may involve following another aircraft or keeping it in sight until it is no longer a factor.

¹⁵ Conflict Alert (CA) - An automated function of the installed radar system at DCA that alerts the controller to an existing or pending situation between tracked targets (known IFR or VFR aircraft) that require the controller's immediate attention/action. Both aural and visual alerts are provided.

2.0 Flight Track Surveillance Data

Automatic Dependent Surveillance – Broadcast (ADS-B)¹⁶ and radar data¹⁷ were both provided by the FAA. The Standard Terminal Automated Replacement System (STARS) used both data to provide the track information displayed to air traffic controllers via Fusion¹⁸. The accident helicopter [PAT25] was not transmitting ADS-B data during the accident flight and was operating under an existing Department of Defense (DOD) exemption that permitted this. However, both aircraft were tracked via radar, and these data were used in constructing the flight track graphics included in this report. The altitude information in these figures is the same as that displayed to air traffic controllers in the tower and are encoded in 100-foot increments.

NOTE: All altitudes, regardless of source, are rounded to the nearest 100-foot increment for display to ATC for efficient communications, airspace management, and consistent separation of air traffic.

Figures 1-2 are overhead views of the entire flight tracks of both the accident aircraft and helicopter separately. Figures 3-5 are profile views of the final segment of the flight tracks of the accident aircraft and helicopter together. All flight track information has been overlaid onto satellite imagery in Google Earth.

¹⁶ ADS-B data provided by the FAA from the CRJ [JIA5342] is included in Attachment 4: ADS-B Data.

¹⁷ Radar data from the STARS at PCT TRACON was provided by the FAA is included in Attachment 5: STARS Radar Data.

¹⁸ According to the FAA, Fusion is the combination of all available surveillance sources (airport surveillance radar [ASR], air route surveillance radar [ARSR], ADS-B, etc.) into the display of a single tracked target for use by air traffic control in providing separation services.

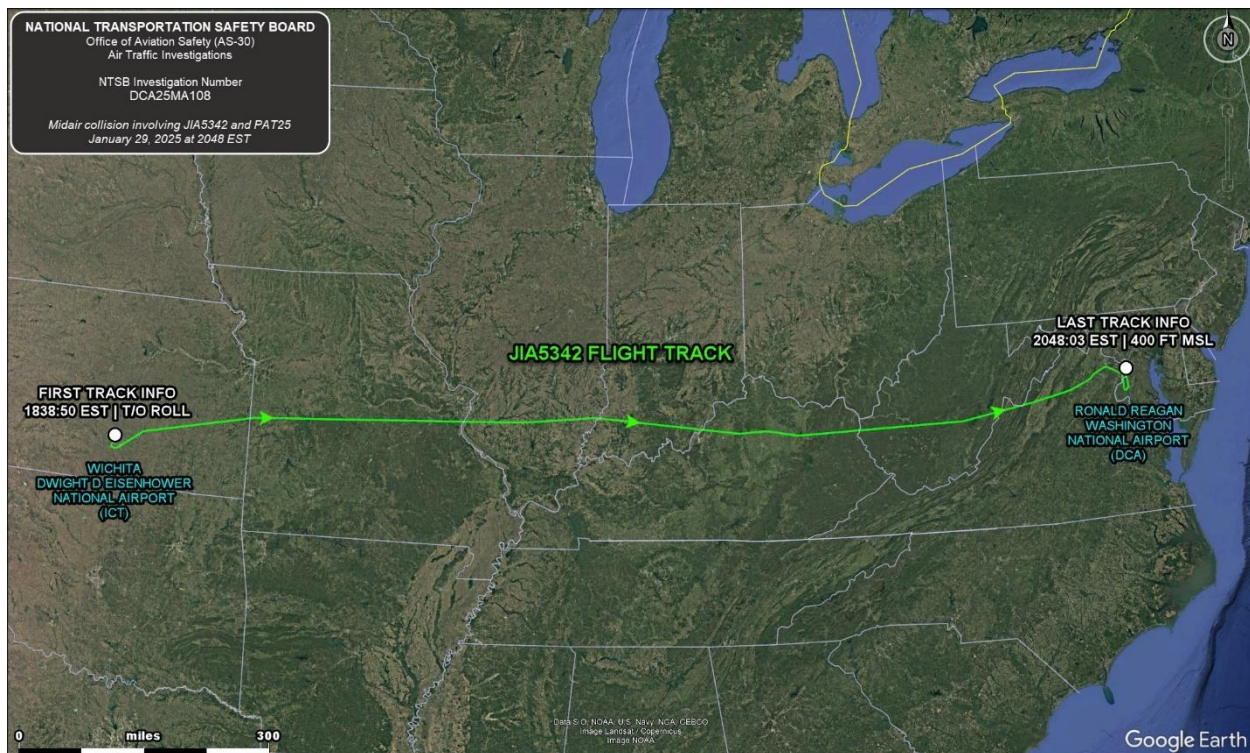


Figure 1. Overhead view of the entire accident flight track of JIA5342 overlaid onto satellite imagery in Google Earth.

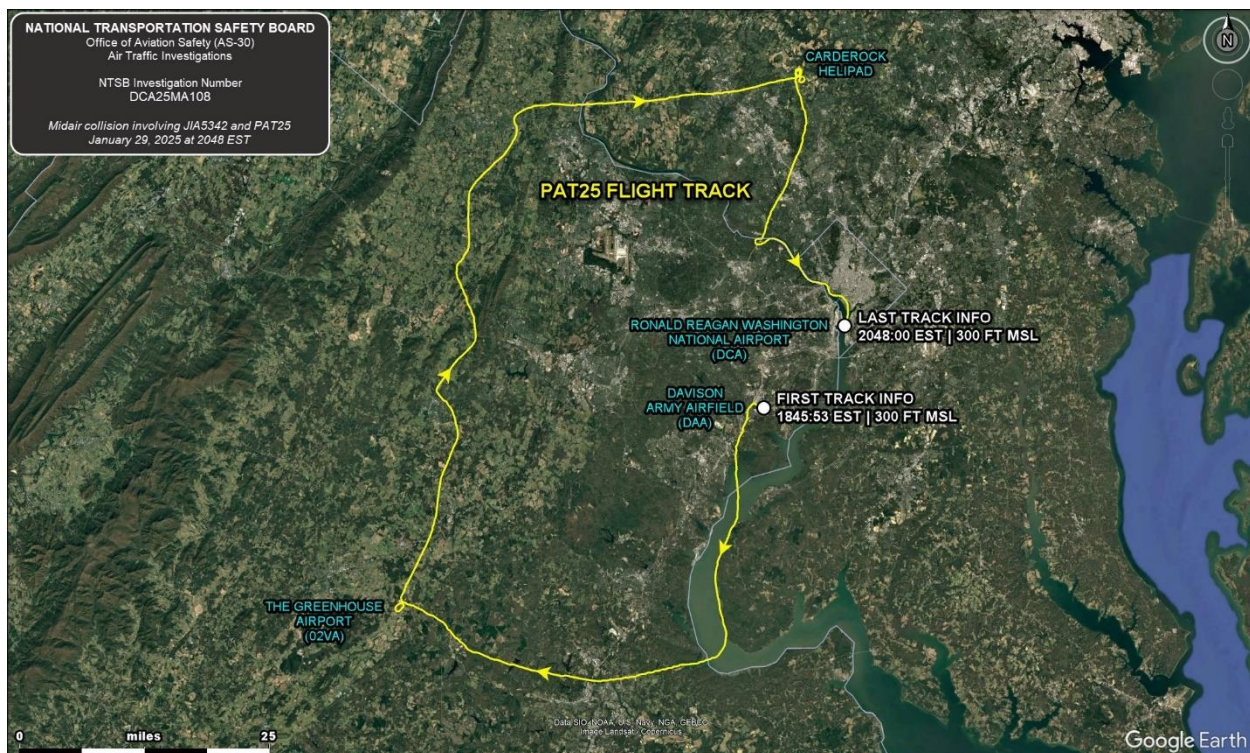


Figure 2. Overhead view of the entire accident flight track of PAT25 overlaid onto satellite imagery in Google Earth.

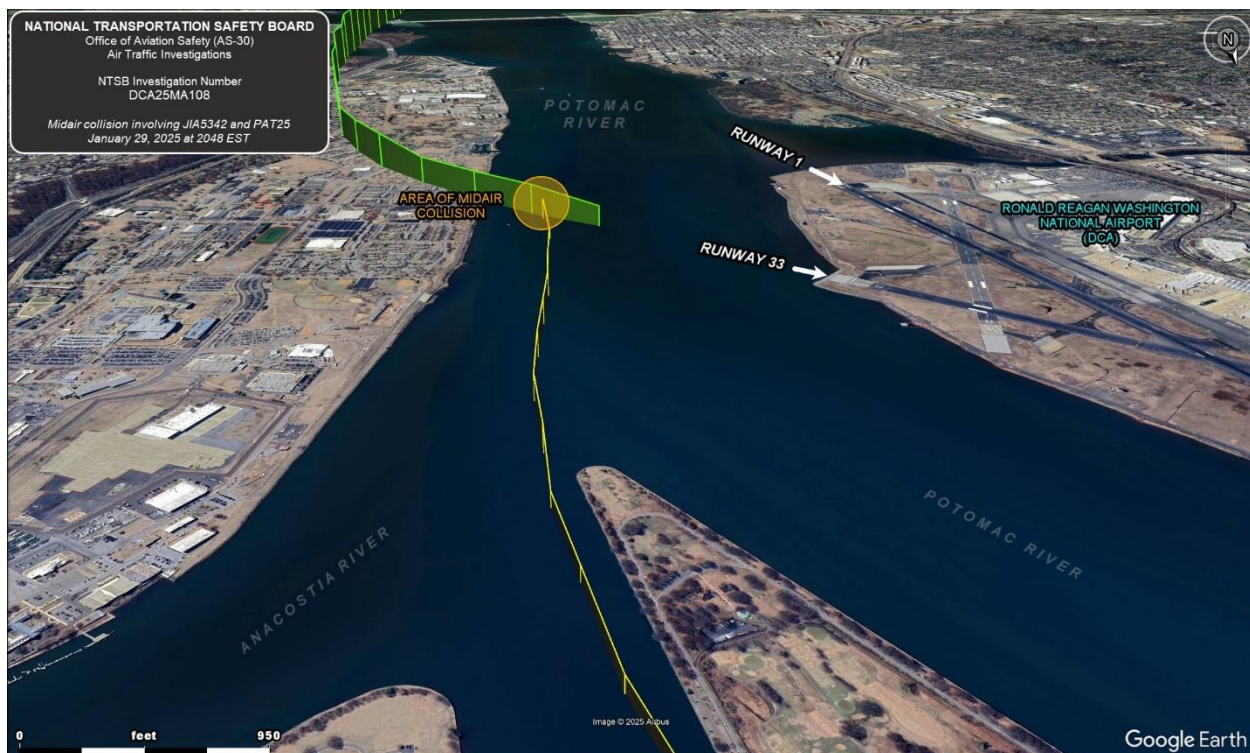


Figure 3. Profile view of the final segment of the accident flight tracks of JIA5342 and PAT25 as viewed from the general direction PAT25 was flying.

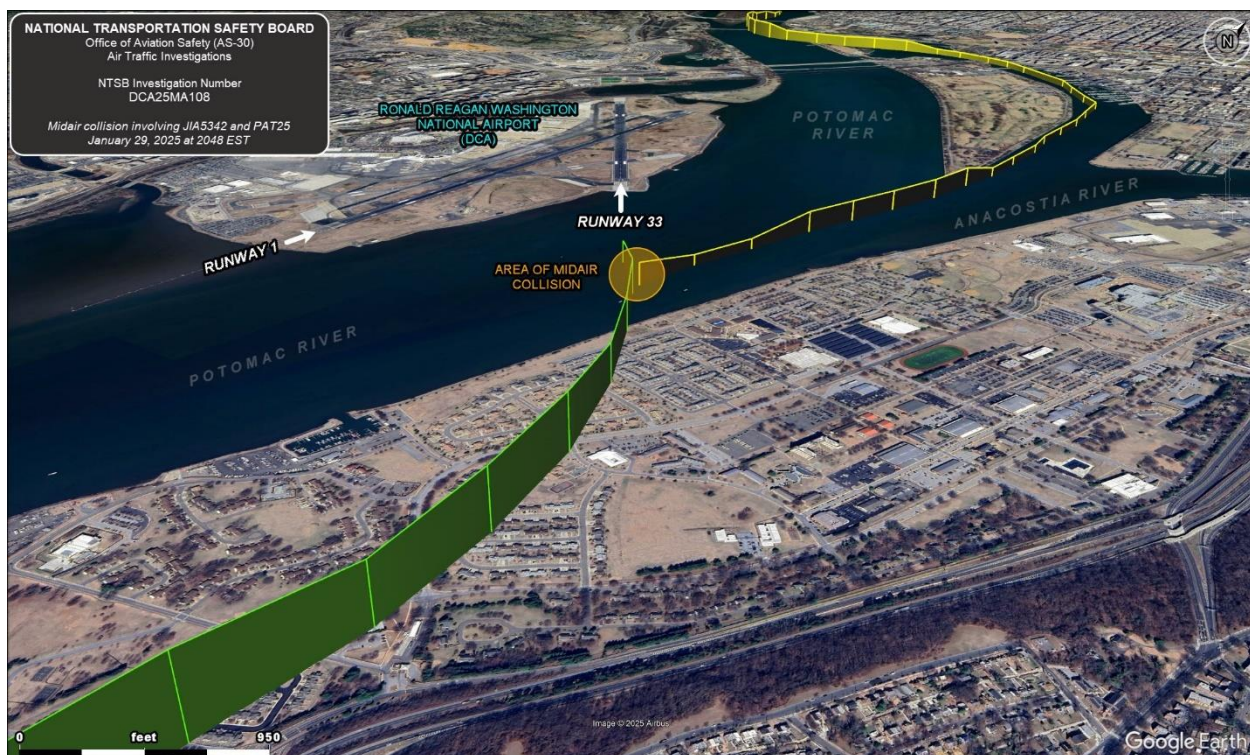


Figure 4. Profile view of the final segment of the accident flight tracks of JIA5342 and PAT25 as viewed from the general direction JIA5342 was flying.

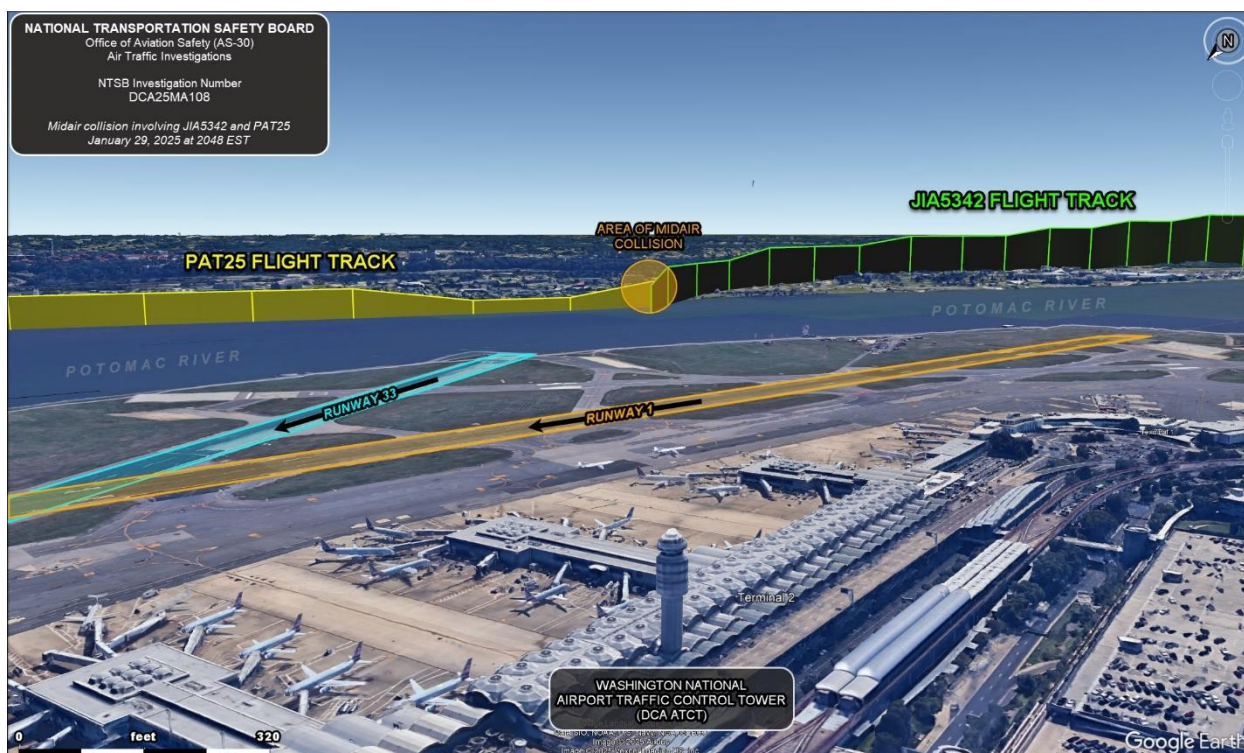


Figure 5. Profile view of the final segment of the accident flight tracks of JIA5342 and PAT25 as viewed from the general direction of the control tower viewpoint.

3.0 Weather Information

At an elevation of 14.1 feet, DCA had an Automated Surface Observing System (ASOS)¹⁹ which was augmented by certified weather observers. The latest sectional chart for the area indicated a magnetic variation of 11° W over the area. DCA was reporting visual flight rules (VFR) conditions, and the following surface observation was disseminated longline²⁰ on weather circuits. For detailed weather information, see the Meteorology Factual Report in the docket.

3.1 Surface Observation

At 2052, DCA reported wind from 300° true at 14 knots with gusts to 23 knots, wind direction variable from 270° to 330°, visibility of 10 statute miles or greater, sky

¹⁹ ASOS – Automated Surface Observing System – Automated sensor suites equipped with meteorological instruments to observe and report wind, visibility, ceiling, temperature, dewpoint, altimeter, and barometric pressure. These systems generally report at hourly intervals but also report special observations if weather conditions change rapidly and cross aviation operation thresholds.

²⁰ "Longline" refers to the dissemination of weather observations with the intent that they are available in near-real time to national databases (effectively, the whole world) and accessible to the general public from a large number of vendors. This does not include public accessibility to observations from a reporting station's Very High Frequency (VHF; line-of-site) or telephone broadcast, where applicable. Longline dissemination of weather observations is the primary vehicle through which the general public has access to surface weather observations, particularly outside of the aviation community.

clear, temperature of 10° Celsius (C) and dew point temperature of -7° C, altimeter setting of 29.90 inches of mercury; Remarks: automated station with a precipitation discriminator, peak wind of 33 knots from 300° true occurred at 2008 EST, sea level pressure of 1012.6 hectopascals, temperature of 10.0° C and dew point temperature of -7.2° C.

VFR METAR KDCA 200152Z 30014G23KT 270V330 10SM CLR 10/M07
A2990 RMK AO2 PK WND 30033/0108 SLP126 T01001072=

4.0 Performance Data Analysis and Reporting System Studies

According to ATAC²¹, Performance Data Analysis and Reporting System (PDARS) had continuously collected flight plan and radar track data since the initial prototype was deployed in 1999. This information came from systems at ARTCCs, which track and provide service to an aircraft for their entire flight, and at the TRACON facilities, which track and provide service to aircraft approaching and departing between 5 and 50 miles of an airport, and most recently from ATCTs, which track and provide service to aircraft on the airport surface and immediate vicinity.

It could look at single days or multiple days, limited only by the available data. PDARS had many applications for ATM safety assurance, including airspace optimization, incident analysis, impact analysis of temporary airspace modifications, etc. Visualization tools in PDARS were extremely suitable to support qualitative analysis. PDARS did not require any set level of data, provided results with little available data and better results for large data sets. Specific goals of PDARS included:

- Collect and manage data
- Process data and generate reports daily
- Share data and reports among facilities
- Support exploratory and causal analysis
- Archive basic operational data and measurements

There were three separate PDARS studies conducted following and related to this accident that the ATC/HP group was aware of. The first was requested by the FAA and released on February 1, 2025, with at least six subsequent updates made as of June 1, 2025. The second was requested by the NTSB ATC/HP group on February 25, 2025, and was not released until April 22, 2025, because of a priority request made by the FAA deputy director for safety and technical training (AJI-1). This was the third request and was provided to the FAA sometime between mid-March and mid-April 2025. The ATC/HP group requested a copy of the first from the FAA party coordinator and was shared with the NTSB on-scene. After repeated requests for the

²¹ ATAC is the primary contractor to the FAA supporting the PDARS program. ATAC's role includes systems engineering, software development and deployment, system monitoring, training, and user support.

third study however, AJI-1 declined to release this study to the NTSB, and instead provided version 7 of the first study which was not responsive to the request.²²

At the time of this writing, the group continued their review and analysis of these data. In accordance with NTSB/FAA Joint Policy Letter dated June 4, 2014 (also referred to as the "Ashburn Accord")²³, the NTSB reiterated our request for a copy of the third study. As the NTSB continues our investigation in partnership with the FAA, these data can help significantly in determining the most accurate probable cause and are critical in developing the most meaningful safety recommendations to industry stakeholders.

Figure 6 and Table 1 below illustrate some of the information that was included in these studies. Figure 6 is an excerpt from the first study and is a comparison of military vs non-military helicopter flights at DCA from 1/1/2024 to 1/30/2025. The flight tracks are in white, and the runways and helicopter routes have been colored and included for reference. Table 1 is an excerpt from the second study and shows the average miles in trail (MIT) being delivered for aircraft landing runways 1 and 33. The MIT values were not computed "to the deck" as required for ATC spacing requirements, but rather to BADDN, which is an airspace fix and reporting point charted on the Mount Vernon Visual Runway 1 approach at 5.9 distance measuring equipment (DME).

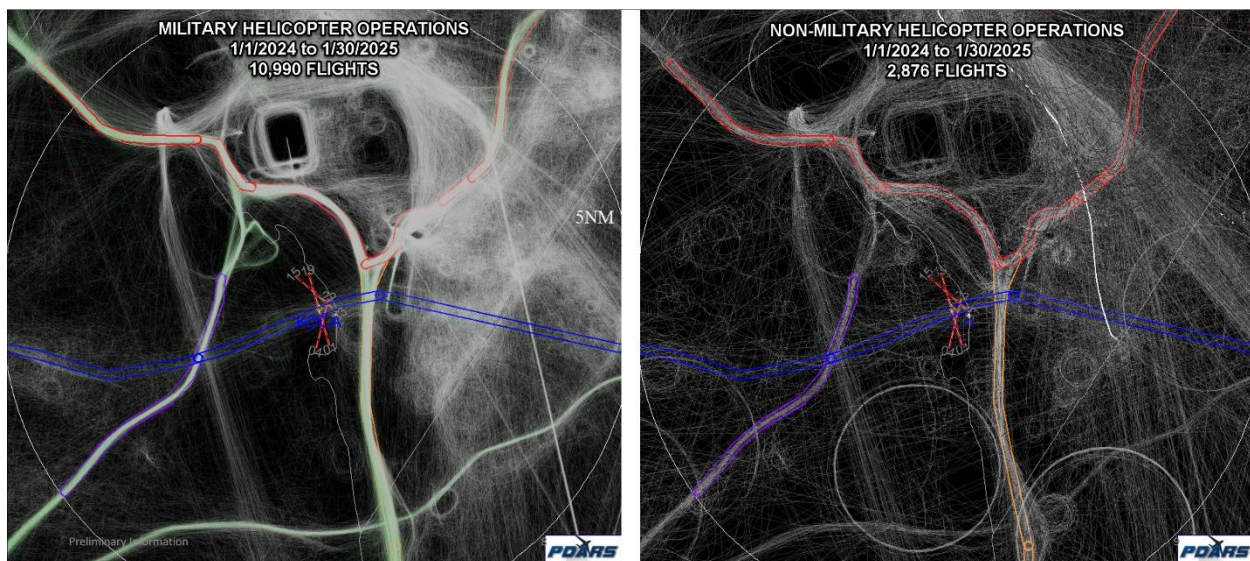


Figure 6. Side by side comparison of total military (left) vs non-military (right) helicopter flight tracks along the DCA helicopter routes from January 2024 - January 2025. The white lines that appear like "cobwebs" are actual flight tracks.

²² Version 7 of the first PDARS study, as well as the second PDARS study are included in Attachment 6: PDARS Studies.

²³ Copy of the NTSB/FAA Joint Policy Letter is included in Attachment X: NTSB/FAA Joint Policy Letter

Table 1. Table breaking down miles in trail (MIT) at BADDN for 2018, 2019, 2023, 2024, and 2025 to date. 2020-2021 data was omitted due to COVID.

Weekday	MIT Dist Bin	2018	2019	2023	2024	2025	Total
Mon	0-2	0	2	0	0	0	2
Tue	0-2	0	0	1	2	0	3
Wed	0-2	1	1	1	0	0	3
Thu	0-2	4	1	0	3	0	8
Fri	0-2	1	4	1	0	0	6
Sat	0-2	1	0	0	1	0	2
Sun	2-4	980	656	985	1,036	112	3,769
Mon	2-4	1,116	876	1,136	1,289	129	4,546
Tue	2-4	1,224	979	1,404	1,143	133	4,883
Wed	2-4	1,121	1,026	1,447	1,043	220	4,857
Thu	2-4	1,344	856	792	1,183	96	4,271
Fri	2-4	1,148	921	1,053	1,157	130	4,409
Sat	2-4	585	402	495	676	51	2,209
Sun	4-6	6,711	4,273	7,119	7,181	1,260	26,544
Mon	4-6	7,496	5,705	7,838	8,078	1,002	30,119
Tue	4-6	7,018	5,525	8,684	6,862	1,795	29,884
Wed	4-6	6,634	5,011	7,824	6,702	1,772	27,943
Thu	4-6	7,480	5,310	5,648	8,146	1,211	27,795
Fri	4-6	6,892	5,516	7,443	7,343	1,727	28,921
Sat	4-6	4,593	2,896	4,750	5,069	559	17,867
Sun	6-8	1,973	1,353	1,803	1,691	371	7,191
Mon	6-8	2,081	1,627	1,534	1,634	229	7,105
Tue	6-8	1,935	1,367	1,771	1,577	442	7,092
Wed	6-8	1,608	1,240	1,690	1,370	472	6,380
Thu	6-8	2,143	1,566	1,228	1,814	274	7,025
Fri	6-8	2,004	1,539	1,674	1,654	390	7,261
Sat	6-8	1,759	1,148	1,583	1,583	222	6,295

F. WASHINGTON NATIONAL AIRPORT TRAFFIC CONTROL TOWER

DCA ATCT was a level 9 tower ATC facility located in Arlington, VA, along the west bank of the Potomac River. It was located within the Washington Tri-Area Class B airspace, which was also in the DC Special Flight Rules Area (SFRA). According to information provided by the Metropolitan Washington Airport Authority (MWAA), the tower stood about 210 feet tall. It was located on the northwest quadrant of the airport, co-located with terminal 2 between concourses C and D. Figure 7 below is a photograph of DCA ATCT.



Figure 7. Photograph of the DCA ATCT.

1.0 Limited Radar Services

The control tower radar displays (CTRD) in DCA ATCT were certified and authorized controllers to provide limited radar services to aircraft operating within their delegated airspace as follows:

- Determining an aircraft's identification, exact location, or spatial relationship to other aircraft.
- Providing information, instructions, and radar traffic advisories to aircraft operating within the airspace delegated to DCA.

- Ensuring separation between aircraft within the airspace delegated to DCA Tower.
- Vectoring VFR/SVFR aircraft within the DCA Class B that are radar identified.
- identified.

2.0 Airspace Surrounding DCA

The airspace surrounding DCA was complex and contained many layers and restrictions. It contained three large airports, military airfields, prohibited area 56 (P-56) A and B, special flight rules area (SFRA), flight restricted zone (FRZ), and extensive charted helicopter operating areas and routes supporting the President, VIP, military, law enforcement, and local hospitals. This combination of variants adds considerable complexity to operations and the provision of ATC services - especially at DCA ATCT. The following subsections breaks down the airspace for better understanding. Figure 8 below illustrates the complexity of this airspace as depicted on a VFR sectional chart.

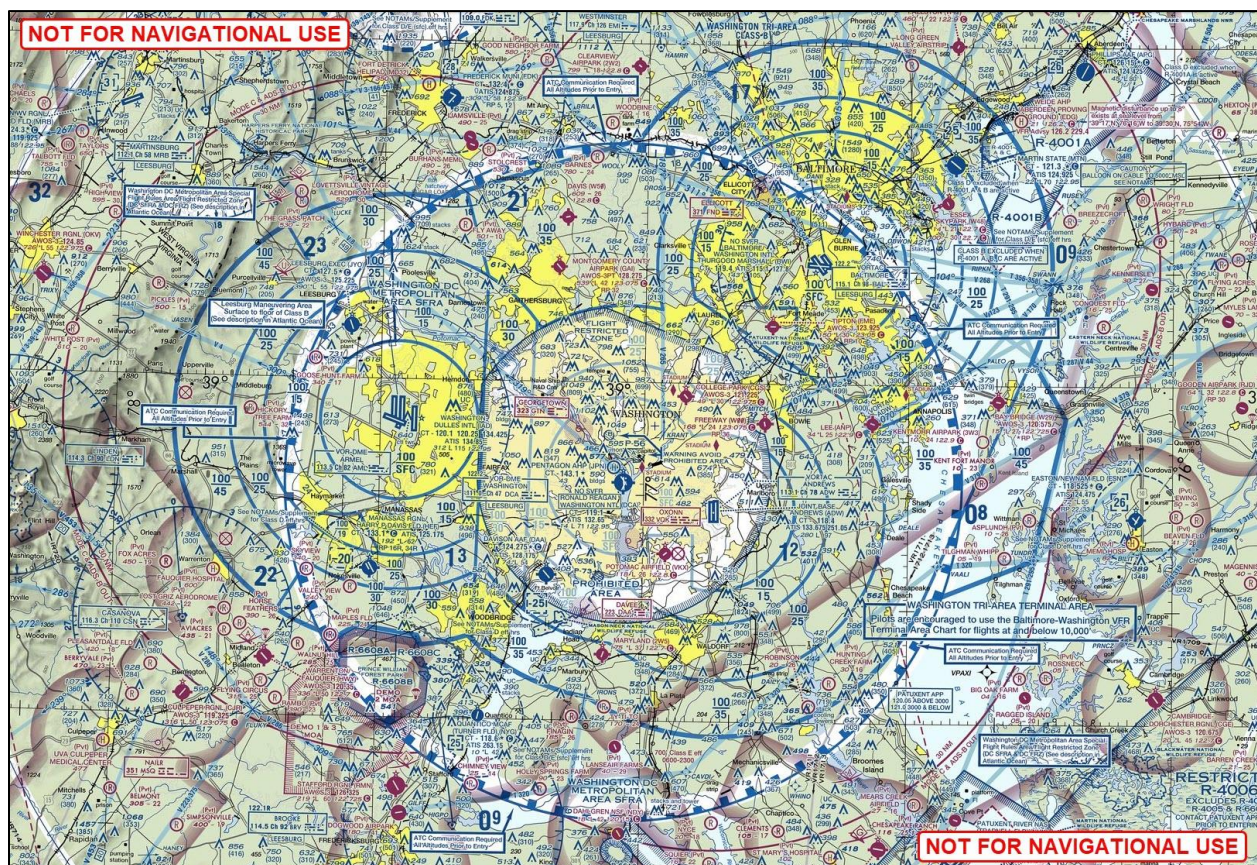


Figure 8. Excerpt from a VFR sectional chart showing the combined airspace layers and boundaries around DCA for reference.

2.1 Washington Tri-Area Class B Airspace

The Washington Tri-Area class B airspace included a surface area, multiple layers, and the DC SFRA, which included the DC FRZ for national security purposes. Figure 9 below illustrates the many layers and boundaries included within the class B airspace surrounding DCA.

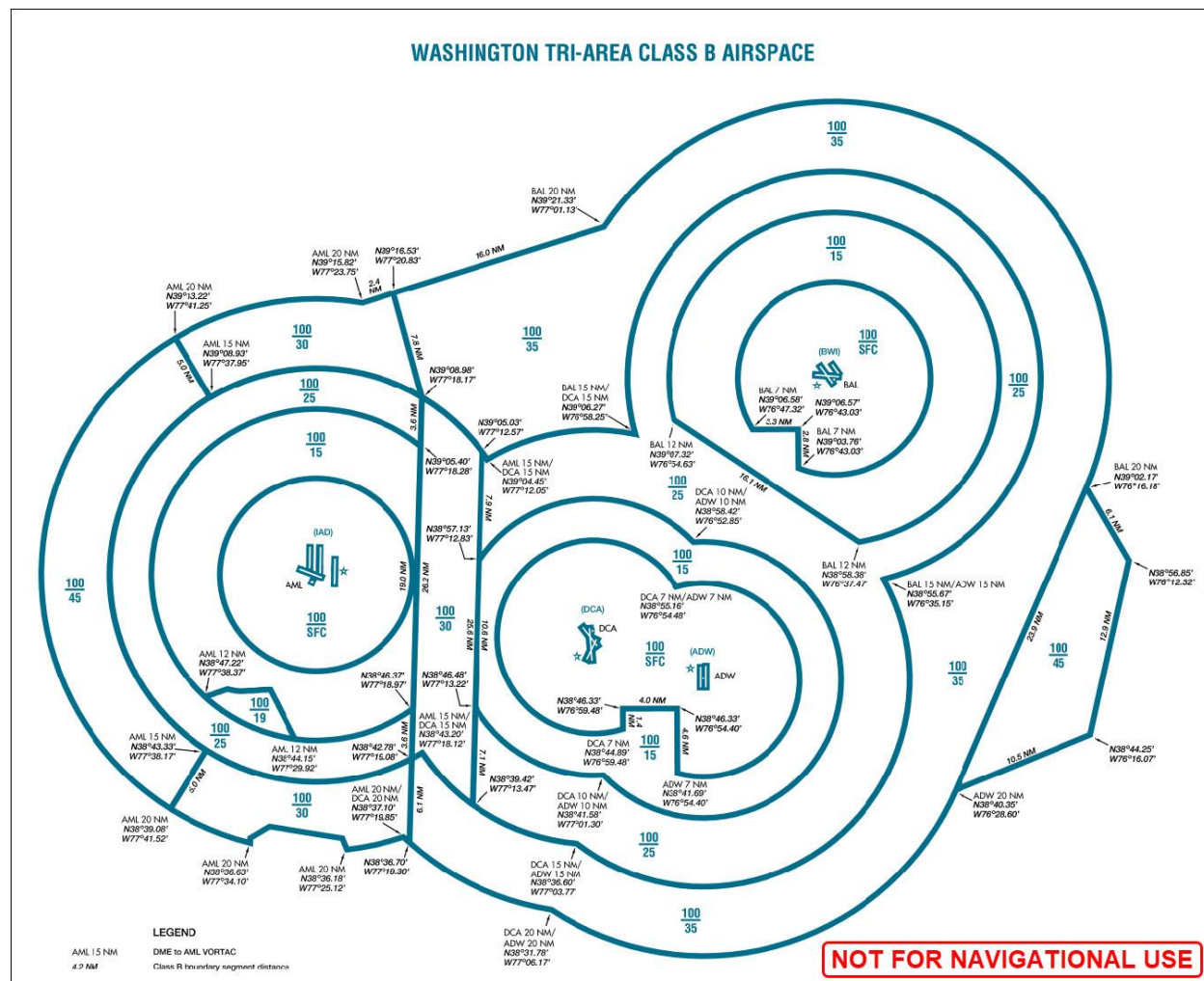


Figure 9. Graphic showing the Washington Tri-Area Class B Airspace.

2.2 Washington DC Special Flight Rules Area (SFRA)

The DC SFRA is an area of airspace over the surface of the earth where the ready identification, location, and control of aircraft is required in the interests of national security. Specifically, the DC SFRA is that airspace, from the surface to, but not including, FL 180, within a 30-mile radius of the DCA VOR/DME. The DC SFRA included the DC FRZ. Figure 10 below shows the DC SFRA and FRZ boundaries.

2.3 Washington DC Flight Restricted Zone (FRZ)

The DC FRZ extends approximately 15 nautical miles around DCA. The only non-governmental flights allowed within the FRZ without a waiver are scheduled commercial flights into and out of DCA. Airlines operating charter flights that support the U.S. government may land at Joint Base Andrews Air Force Base (ADW) or DCA airports without a waiver and under certain conditions per FDC NOTAM 8/3032.

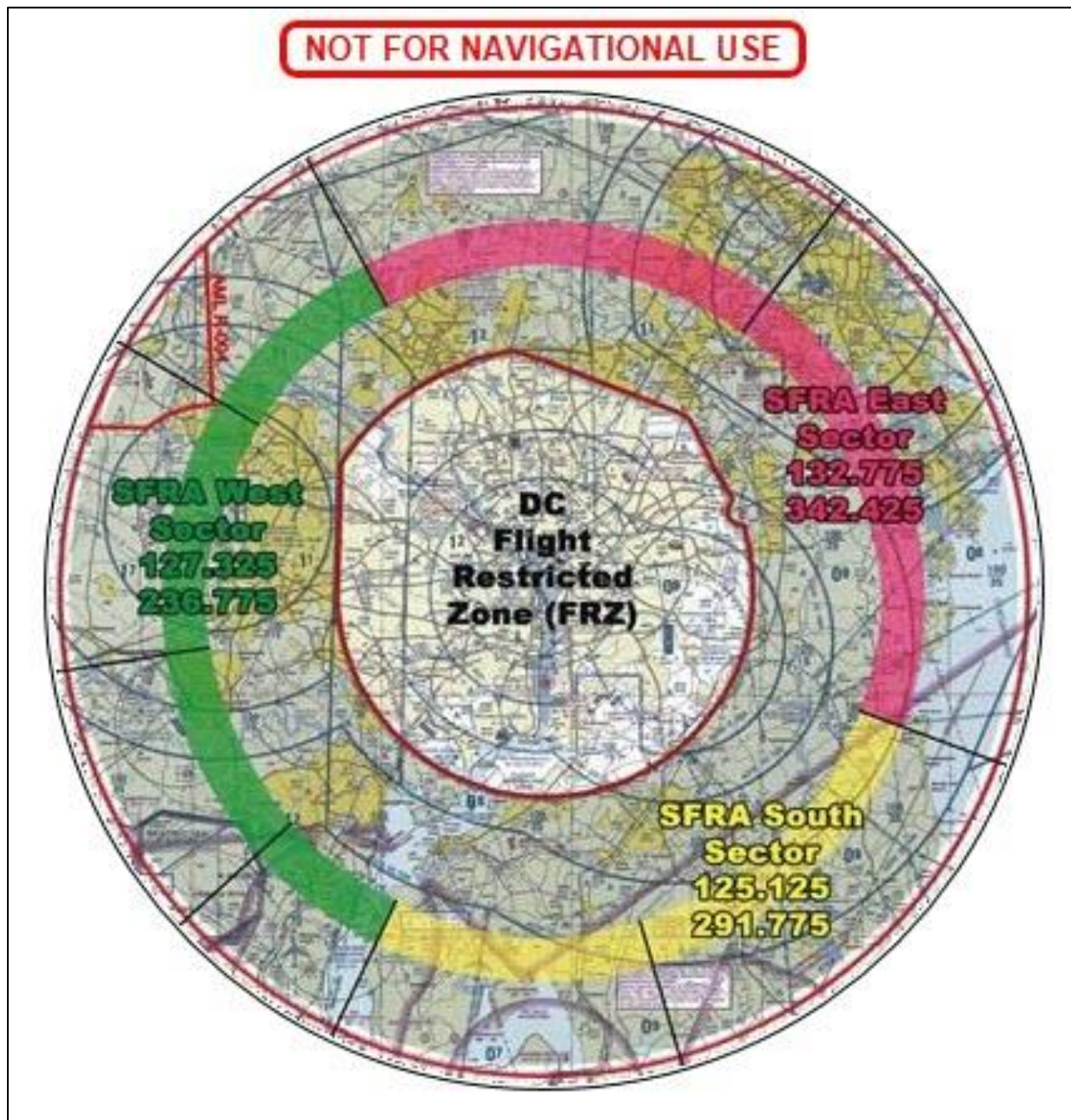


Figure 10. Sectorized depiction of the DC SFRA and DC FRZ.

2.4 Prohibited Area 56 (P-56)

P-56A and B are prohibited areas surrounding the White House, the National Mall, and the Vice President's residence in Washington, DC, only a few miles NNW and NNE of DCA. The only aircraft that are allowed to fly within these prohibited areas are specially authorized flights that are in direct support of the US Secret Service, the Office of the President, or one of several government agencies with missions that require air support within P-56.

2.5 DCA Area Helicopter Operations

The DCA class B airspace contains a high volume of helicopter activity from a myriad of operators including, but not limited to, local and state law enforcement, all branches of the military, Homeland Security, Federal Bureau of Investigation (FBI), National Park Service, medical transport, and flight schools. This volume and complexity of mixed operations required letters of agreement²⁴ with many of the operators that outline specific procedures while operating within the DCA class B.

2.5.1 Baltimore-Washington Helicopter Routes

The group met with an aeronautical information specialist and group manager from FAA AJV-A2 to better understand the history and current information contained on the Baltimore-Washington Helicopter Route chart.²⁵ The specialist explained that helicopter routes as depicted on these charts had no lateral boundaries and were meant to follow linear features such as highways or shorelines. The lines drawn on the chart depicting a helicopter route are not scaled and have no associated measurements.

While they could not recollect the exact history, they stated that the routes had mostly been developed decades ago, with minimal changes over the years, mostly involving altitude changes. They stated that when a helicopter route was submitted for inclusion in charting, the desired routes were developed by ATC and a proposal would be sent to their office. From there, a prototype chart would be developed and provided back to the requester for review. Once agreed that the charting met the needs of the request, it would then be finalized for publication. They stated that there had been no changes to this chart between 2014 and June 2021, when the chart was updated with an altitude change for route 1, changing the altitude from 200 feet to 300 feet near the South Capitol Street Bridge.

²⁴ All LOAs that were current at the time of the event are included in Attachment 7: DCA ATCT Letters of Agreement.

²⁵ The conversation held with the FAA employees from AJV-A2 was documented in a Memorandum of Record and is included in Attachment 8: Memorandum of Record - AJV-A2.

They stated that to their knowledge, ATC conducted annual reviews of these charts and the associated airspace and would submit any required changes to their office. They explained that FAA Order JO 7210.3DD outlined the procedures for requesting changes to aeronautical charting. Figure 11 below is an excerpt from the Baltimore-Washington Helicopter Route Chart that was valid at the time of the accident.

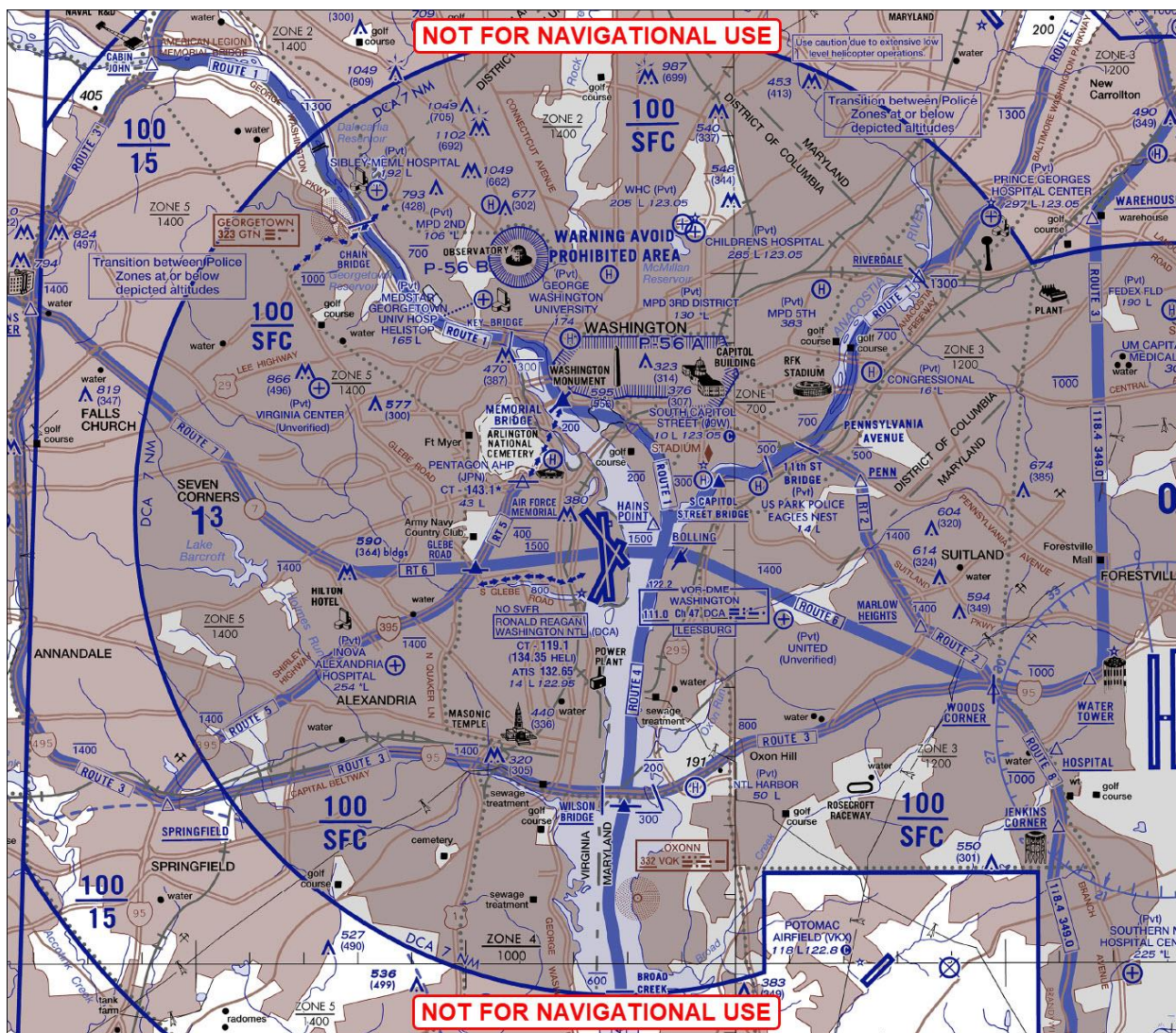


Figure 11. Excerpt from the Baltimore-Washington Helicopter Route Chart that was valid at the time of the accident.

In the weeks immediately following the accident, after urgent safety recommendations (A-25-001 and A-25-002) were issued by the NTSB, the FAA initiated a change to the helicopter routing structure, eliminating the portion of helicopter route 4 between Hains Point and the Wilson Bridge. According to chart update effective June 12, 2025, helicopter route 4 had been eliminated altogether. Figure 12 below is an excerpt from the updated chart.

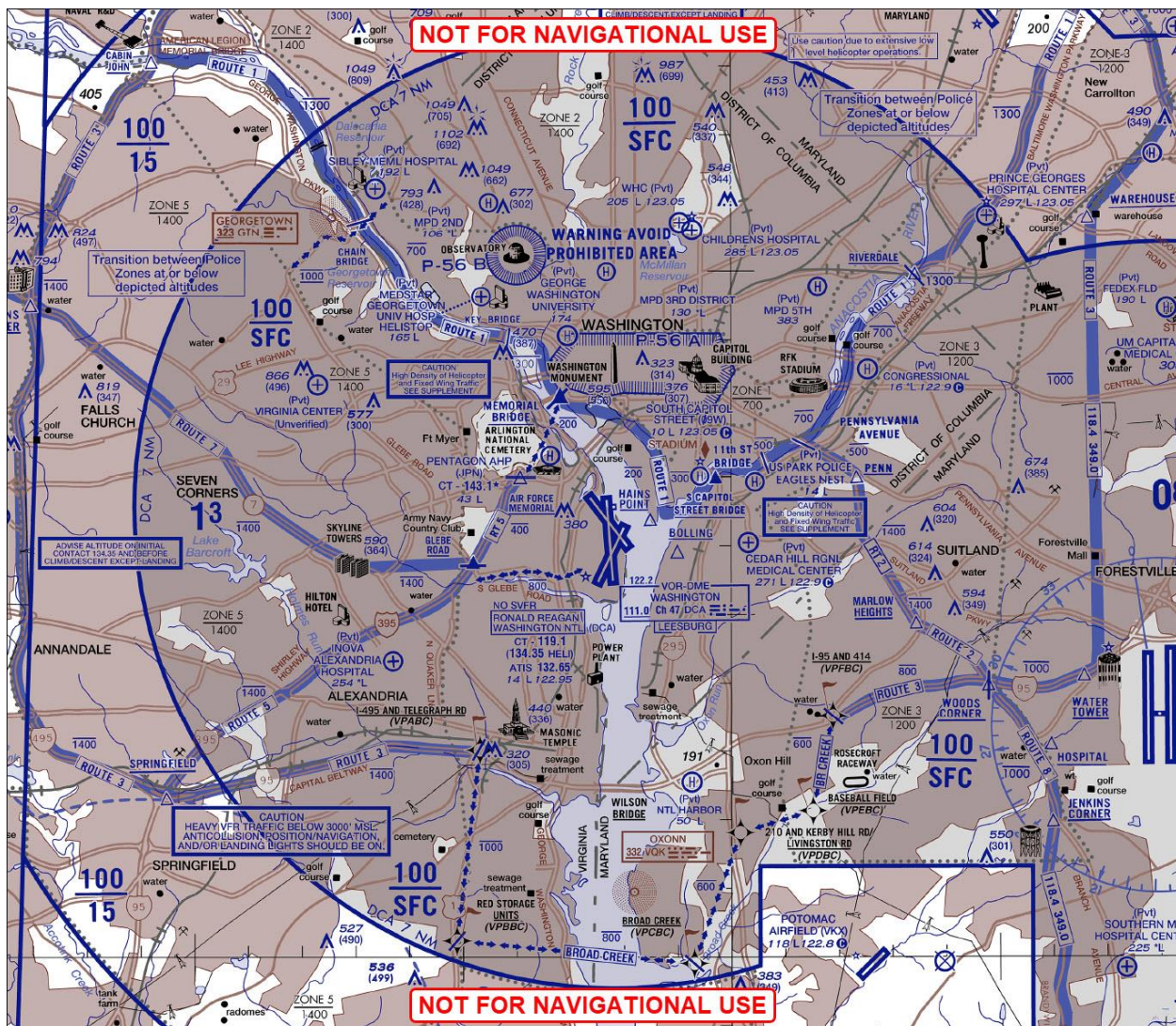


Figure 12. Excerpt of the Baltimore-Washington Helicopter Chart effective June 12, 2025.

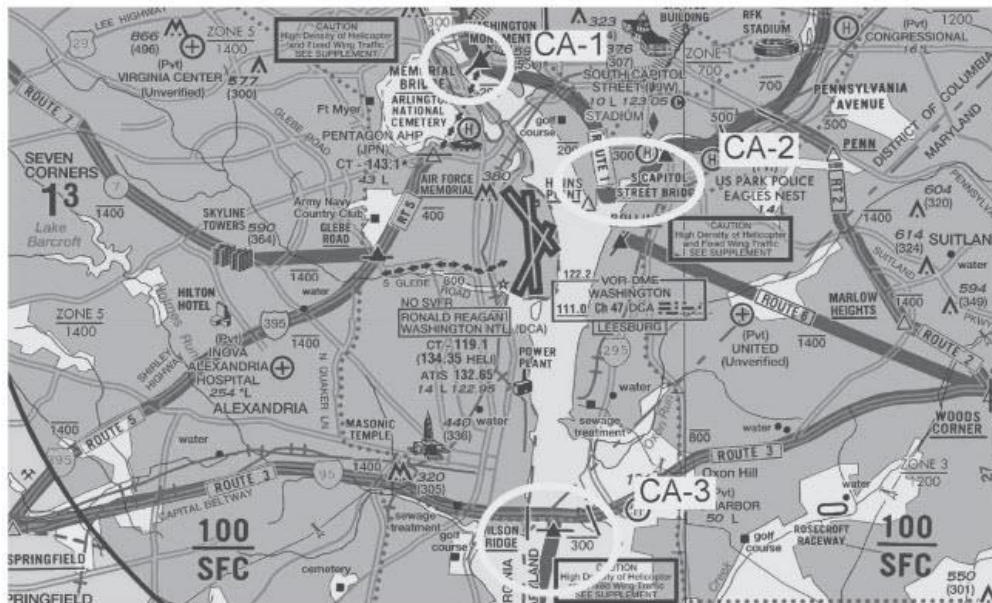
Additionally, along with the change to the helicopter route structure and associated charting, the FAA released a Special Notice in the DCA Chart Supplement. This Special Notice published three "Cautionary Areas" to illustrate areas of concern to airmen. The notice warned airmen of the high concentration of helicopter and fixed wing traffic and IFR jet operations below 1,500 feet. It reiterated requirements of aircraft operating within the class B airspace, provided specific alerts for various helicopter routings/transitions, and lastly it included a chart depicting the three cautionary areas by overlaying on an excerpt of the Baltimore-Washington Helicopter Route chart. Figure 13 below is a copy of the Special Notice from the DCA Chart Supplement that was effective April 17, 2025.

**CAUTIONARY AREA (CA)
HIGH DENSITY OF HELICOPTER AND FIXED WING TRAFFIC
Ronald Reagan Washington National Airport (DCA)**

Pilots should be alert in this area due to heavy concentration of IFR Jet Operations below 1500 Feet. On initial contact with DCA Tower (134.35/257.6), pilots must announce intended route and receipt of DCA ATIS Information (132.65).

- Heavy concentration of helicopter and fixed wing traffic. Pilots should avoid using Route 1 South of the Memorial Bridge to the maximum extent possible.
- All aircraft operating within the DCA Class B are required to:
 - Turn-on Anti-collision lights and aircraft position/navigation lights. Use of landing lights is recommended.
 - Have a current Baltimore-Washington Helicopter Route Chart in the aircraft and familiarize themselves with the information contained therein.

NOT FOR NAVIGATIONAL USE



CA-1 : JPN/Memorial Bridge Transition

- Be alert for RWY 1/33 departures when DCA is in a North Configuration and RWY 15/19 arrivals when in a South Configuration.
- Be alert for JPN (Pentagon) helicopter traffic and route/zone transition area.

CA-2 : Hains Point/South Capitol Region

- Be alert for helicopters operating in/out and local pattern of Joint Base Anacostia-Bolling, South Capitol Street Heliport, US Park Police Eagles Nest.
- Be alert for RWY 4 departures when DCA is in a North Configuration.

CA-3 : Wilson Bridge/Zone 4

- Be alert for fixed wing arrivals to RWY 1 and circling to RWY 33.
- Be alert for Route 3, Route 4 and Zone 4 converging helicopter traffic.
- Do not loiter in Zone 4 over the Potomac River.

Office of Primary Responsibility (OPR): Washington National ATCT (DCA)
Contact Information: 703-413-0336; email: 9-AJO-DCA-OPS@faa.gov
Original: March 2025

Figure 13. Excerpt of the Special Notice from the DCA Chart Supplement regarding high density helicopter and fixed wing traffic.

3.0 Tower Cab Layout

The DCA ATCT faced east looking out across the Potomac River, and on the day of the accident, DCA was in a "North Operation," which meant that they were landing and departing runway 1, with intermittent arrivals to runway 33. Figure 14 below is a diagram of the tower cab layout, and the positions have been labeled as they were staffed at the time of the accident. Additionally, though not staffed separately at the time of the accident, the HC position has also been labeled for awareness.

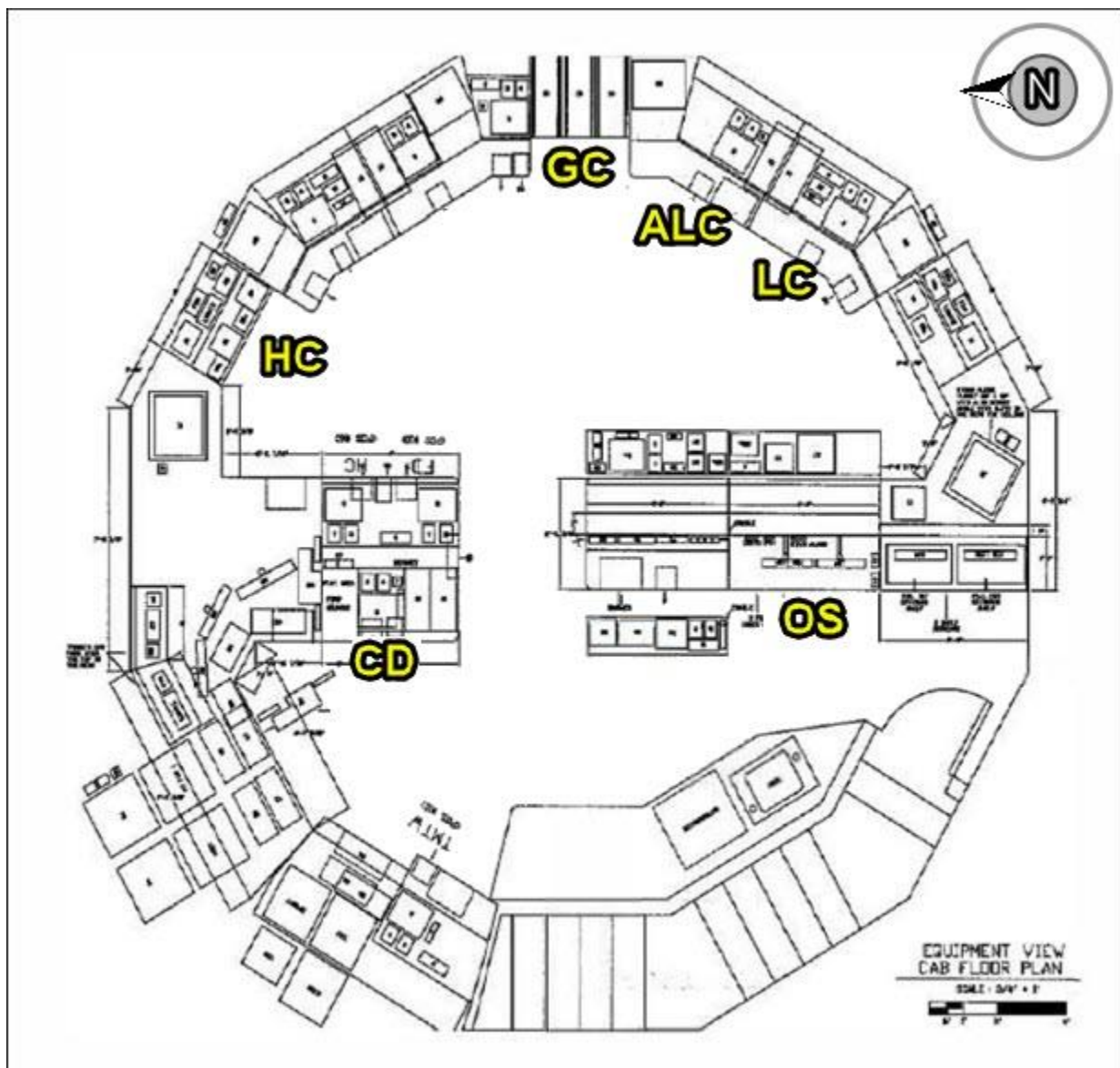


Figure 14. DCA ATCT tower cab layout diagram.

4.0 Pertinent Tower Equipment

DCA ATCT was equipped with numerous systems for use in providing ATC services. Below are some of the more pertinent installed equipment that was available to controllers on the evening of the accident. Controller workstations were individually documented and included in attachment 9 of this report.

4.1 Control Tower Radar Display (CTRD)

The tower was equipped with multiple certified CTRDs. The CTRDs at DCA ATCT were fed by the PCT TRACON STARS radar feed and were used to provide limited radar services to aircraft operating within the DCA class B airspace. The system is configurable and included integrated system safety logic to provide aural and visual alerting to controllers. Conflict alerts were presented to controllers with a visual (red flashing "CA" on the CTRD next to each target involved in the conflict) and an audible alert. Figure 15 below is a photograph of one of the CTRDs installed in DCA ATCT.

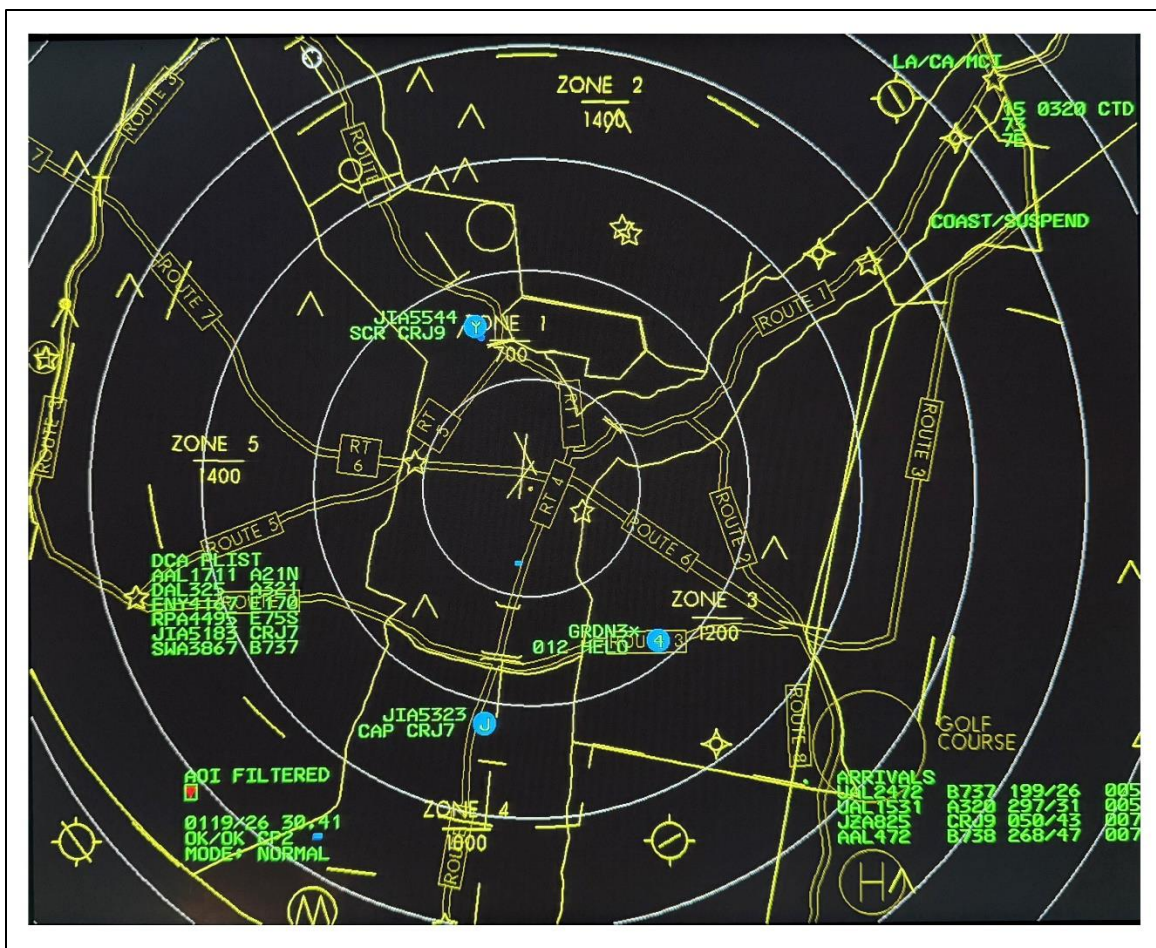


Figure 15. Photograph of one of the CTRDs in DCA ATCT. Displayed information is not related to the accident sequence.

4.2 Airport Surface Detection Equipment - Model X (ASDE-X)

DCA had an installed ASDE-X surface surveillance radar system and the DCA ATCT was equipped with multiple ASDE-X displays mounted at various positions in the tower cab. The ASDE-X provided controllers with a ground radar system capable of tracking aircraft and vehicles on the airport surface. It used radar, multilateration, and satellite technology to provide controllers with real-time data, aural and visual alerting, and enhanced situational awareness. Figure 16 below is a photograph of one of the ASDE-X displays installed in DCA ATCT.

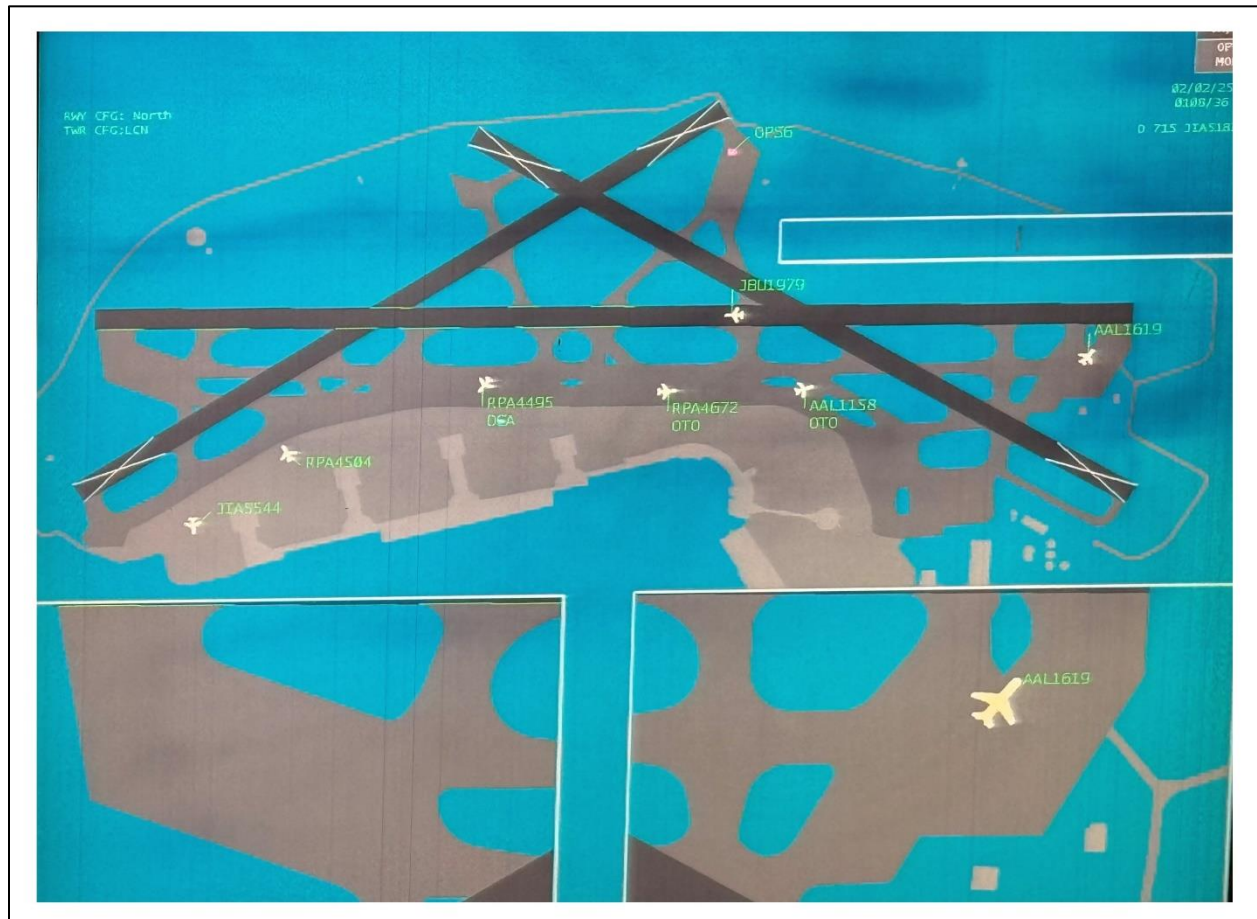


Figure 16. Photograph of one of the ASDE-X displays in DCA ATCT. Displayed information is not related to the accident sequence.

4.3 Information Display System - Version 5 (IDS-5)

DCA ATCT was equipped with the IDS-5 system of workstations that provided controllers with immediate access to relevant operational information and support data that added situational awareness. There were multiple IDS-5 workstations installed in the tower. Information available to controllers included weather information from the airport's ASOS, flight data input output (FDIO), navaid status,

and notices to air missions (NOTAMs). Figure 17 below is a photograph of one of the IDS-5 displays installed in DCA ATCT.



Figure 17. Photograph of one of the IDS-5 displays in DCA ATCT. Displayed information is not related to the accident sequence.

4.4 Enhanced Telephone Voice Switch (ETVS)

DCA ATCT was equipped with an ETVS system for ATC communications and was available at every control position in the tower. The ETVS used a touchscreen console that was configurable at the facility level and then used by ATC to access radio and landline communication networks. This is the primary hub in which all communications from the controller to/from aircraft, other positions within the tower, and adjacent facilities was conducted. Figure 18 below is a photograph of one of the ETVS consoles installed in DCA ATCT.

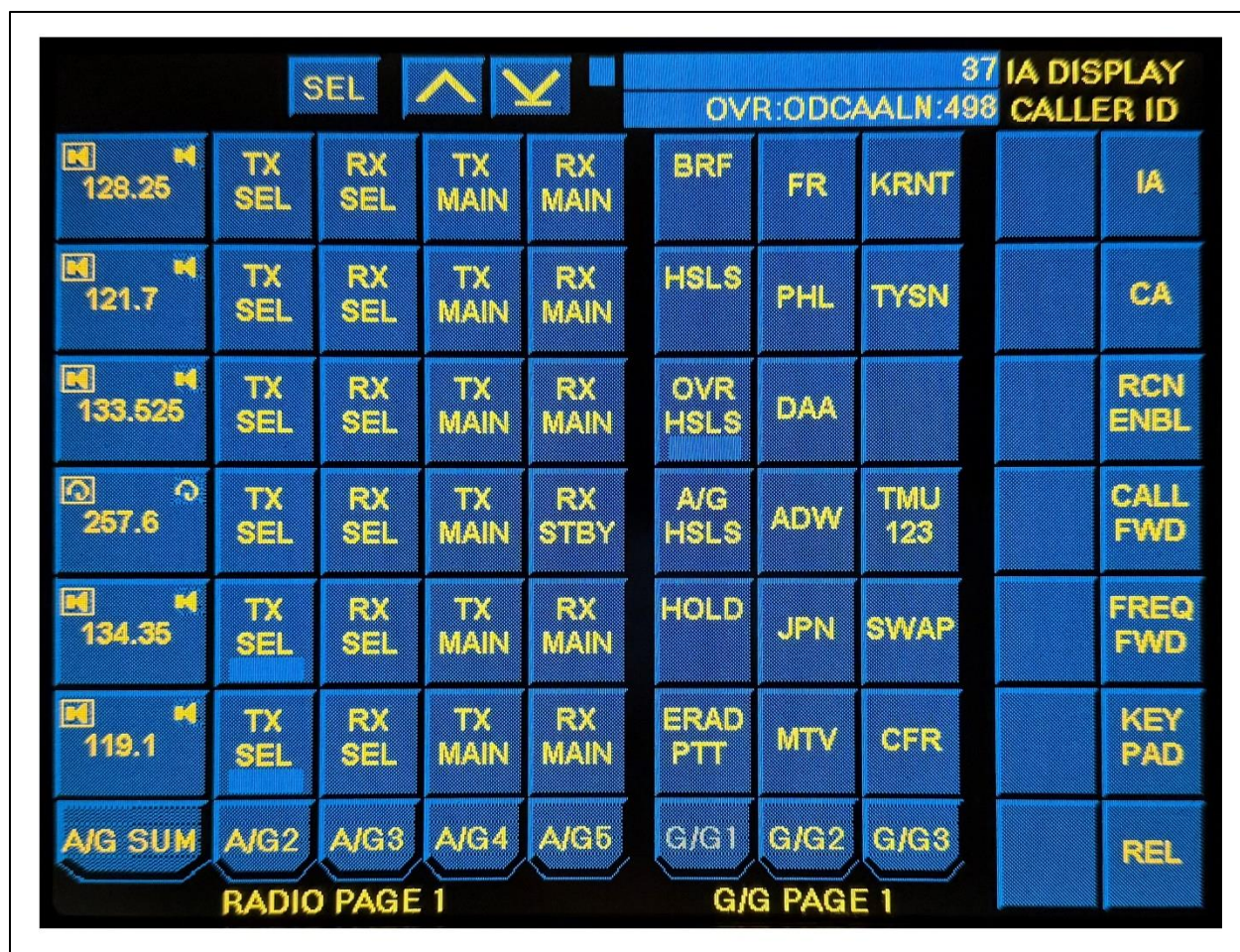


Figure 18. Photograph of one of the ETVS control panels in DCA ATCT. The displayed information is not related to the accident sequence.

5.0 Facility Health

5.1 Staffing

After considerable confusion and a series of corrections and updates from the FAA regarding the staffing numbers at DCA ATCT (both authorized and onboard at the time of the event), the latest clarification²⁶ provided by the newly assigned acting ATM (replacement for the ATM that was in place at the time of the accident) is detailed below. In the following staffing numbers, operational refers to those that were qualified and available to work in the staffing rotation, and temporary refers to those on temporary details awaiting permanent replacements. According to the ATM there were 16 positions per day that need to be staffed. Overtime was unable to be confirmed by the FAA as of June 18, 2025.

²⁶ Staffing numbers provided by the FAA are located in Attachment 10: DCA ATCT Staffing Documentation.

5.1.1 Authorized and Onboard Staffing - January 29, 2025

On January 29, 2025, DCA ATCT was authorized 28 certified professional controllers (CPCs), 3 traffic management coordinators (TMCs), 6 OSs, 1 OM, 1 support manager (SM), 3 staff support specialists (SSS), and 1 ATM. The onboard staffing at that time was 28 (25 operational) CPCs, 2 TMCs, 6 (4 operational) OSs, 1 OM, 1 SM, 3 SSS, and 1 (temporary) ATM.

After the accident, the facility lost 2 operational CPCs, 1 operational TMC, 1 operational OS, and an OS-IT to office of workers' compensation (OWCP) leave with continuation of pay (COP). Several weeks later, 1 CPC and the OS-IT returned to operational duty.

5.1.2 Authorized and Onboard Staffing - June 1, 2025

On June 1, 2025, DCA ATCT was authorized 30 CPCs, 3 TMCs, 8 OSs, 2 OMs, 1 SM, 3 SSS, and 1 ATM. The onboard staffing at that time was 28 (25 operational) CPCs, 2 TMCs, 5 OSs, 1 (temporary) OM, 1 SM, 2 SSS, and 1 (temporary) ATM.

5.1.3 Authorized and Onboard Staffing - June 18, 2025

On June 18, 2025, DCA ATCT was authorized 30 CPCs, 3 TMCs, 8 OSs, 2 OMs, 3 SSSs, and 1 ATM. The onboard staffing at that time was 26 (20 operational) CPCs, 2 (0 operational) TMCs, 5 (4 operational) OSs, 2 (temporary) OMs, 2 (1 temporary) SMs, 2 SSSs, and 1 (temporary) ATM.

5.1.4 Projected Gains and Losses

On June 1, 2025, the facility had 2 permanent CPCs, and 4 temporary CPCs identified inbound, with 4 identified outbound. They had 2 temporary TMCs and 1 permanent OS inbound, with 1 OS identified outbound. Additionally, they had vacancies for between 1-3 permanent OMs, and 1 permanent SSS pending. As of January 18, 2025, some of these numbers may have changed but were unable to be verified at the time of this writing.

5.1.5 Collaborative Resource Workgroup (CRWG)

The 2023 Collaborative Resource Workgroup (CRWG)²⁷ numbers raised the target staffing for DCA ATCT from 28 to 31, however, these were only target numbers established by the CRWG and had not yet been implemented at the time of the accident and is unknown if they will be adjusted as a result of the staff shaping that

²⁷ CRWG - Collaborative Resource Work Group - a workgroup was established to collaboratively address the implementation of the 2023 CRWG CPC Targets in a manner that would prioritize staffing in the most critically understaffed facilities, while balancing the need to continue to approve employee requested reassignments that considered the needs of the FAA and the employee.

has occurred since the accident. The CPC onboard to CRWG target percentage at the time of the event was about 90% and remained the same as of June 1, 2025.

5.2 Event Resource Management

At the time of the accident, there were 5 CPCs, 1 TMC, 1 CPC-IT, 1 OS, and 1 OS-IT working. ALC and GC were standalone, CD was combined with FD and being worked from the CD workstation and LC and HC were combined and being worked from the LC north workstation, and an OS was providing direct supervision from the supervisor's workstation. Of the 9 ATC personnel present in the building, there was 1 CPC on LC/HC, 1 CPC on GC, 1 TMC on ALC, 1 OS-IT on CD/FD, and 1 OS providing direct supervision.

According to interviews conducted with the OS working at the time of the accident, no personnel had been released early from shift, and all personnel that were still on shift were available at the facility. Security card access information was requested from MWAA, however it was inconclusive. According to MWAA officials, a person could enter or exit the various ATC facility access points without swiping their individual card. For example, a person could enter or exit with another person or group, or security personnel could use their access to allow authorized personnel access as well. Though "piggybacking²⁸" was not technically authorized, this rule was not enforced with ATC personnel.

5.2.1 Combined Control Positions

The CD and FD positions were combined in accordance with the DCA ATCT standard operating procedures (SOP)²⁹. The positions had been combined for the entire day and was the normal configuration for DCA ATCT. The CD controller working at the time of the accident had been working the CD/FD positions combined since 2021 EST. This was the normal configuration for DCA ATCT and was a common position configuration across the NAS. The CD/FD positions were not providing services to either of the accident aircraft. According to the SOP, the OS was authorized to combine/de-combine the CD/FD positions at their discretion after considering the following:

- Staffing Levels
- Training requirements
- Weather conditions
- Projected traffic load
- Equipment

²⁸ Piggybacking was the practice of multiple personnel accessing controlled access points under one person's access card swipe.

²⁹ A copy of DCA 7210.2L, DCA ATCT Standard Operating Procedures (SOP) is included in Attachment 11: DCA ATCT SOP - Multiple Versions and Notices.

The LC and HC positions were combined in accordance with the DCA ATCT SOP. The positions had been combined since 1540, and prior to that had only been de-combined that day for a total of 1 hour and 20 minutes. The involved LC controller had been working the positions combined since 1928. The SOP had additional guidance to the OS regarding combining of the HC position. According to the SOP, HC “should normally” be de-combined between the hours of 1000-2130 Monday through Friday. Also, the OS was authorized to combine/de-combine the HC/LC positions at their discretion after considering the following:

- Weather conditions
- VIP movements
- Special helicopter operations
- Training initiatives
- Staffing
- Air carrier traffic volume
- Helicopter traffic volume within the DCA Class B airspace

The OS that was providing direct supervision at the time of the accident had worked three sessions as OS on the day of the accident. One from 1630-1732, then from 1823-2159, and lastly from 2200-2205. In post-event interviews, the OS stated that they had not been the one who had combined the positions initially on that day. However, based on workload and other considerations, never felt there was a reason to de-combine them and therefore left them combined.

5.2.2 Helicopter Control Position Staffing

The HC position staffing had undergone several changes over the years leading up to the time of the accident. According to DCA ATCT facility documentation provided by the FAA³⁰, the following is a timeline of these changes.

June 26, 2016 - DCA ATCT SOP Change Notice

This change specifically implemented the requirement to staff the HC position separately during specific hours on each day of the week. The specific wording stated “*HC must be open...*” during the following times:

- 0800-2000 Monday - Friday
- 0900-1900 Saturday
- 0900-2000 Sunday

Additionally, it stated that the position could only be closed at the direction of the ATM or OM and listed no exceptions to this requirement.

³⁰ Documentation is included in Attachment 11: DCA ATCT SOP – Multiple Versions and Notices.

May 15, 2020 - DCA ATCT SOP - DCA 7110.2J

This SOP reiterated the previously implemented requirements to staff the HC position separately during the same specified hours, and could only be closed at the direction of the ATM or OM with no exceptions to this requirement.

February 9, 2022 - DCA ATCT Internal Memo from ATM to all OS/CIC

This memorandum reiterated the requirement for the HC position to be opened separately during the specific hours on each day of the week. The wording remained *"HC must be open..."* during the same timeframes listed in the June 2016 SOP change. It also re-stated that the position could only be closed at the direction of the ATM or OM and listed no exceptions to this requirement.

May 12, 2022 - DCA ATCT Internal Memo from OM to all OS/OIC

This memorandum reiterated all the same requirements as the June 2016 SOP change and February 2022 memo, including the required timeframes. However, it added an exception that allowed the OS/CIC to close the HC position at their discretion during IFR conditions, and that if it was closed for this purpose, must be logged in the daily facility log.

January 1, 2023 - DCA ATCT SOP - DCA 7210.2K

This updated version of the DCA ATCT SOP implemented a change to the HC position staffing requirements. Specifically, it changed the wording to state *"HC must normally be de-combined..."* during the following times:

- 1000-2300 Monday - Friday
- 1000-1700 Saturday - Sunday

Additionally, it stated that the HC position may be closed at the discretion of the OS/CIC after considering a list of 7 items (see section F, sub-section 5.2.1 above), and that it must be logged in daily facility log.

June 21, 2023 - DCA ATCT SOP Change Notice

This change notice removed only the requirement for the OS/CIC to document in the daily facility log when the HC position was combined/de-combined.

June 1, 2024 - DCA ATCT SOP - DCA 7210.2L

This updated version of the SOP made no substantial changes to the previously implemented policy regarding HC position staffing requirements. It did, however, incorporate the interim change notice that was issued in June 2023.

5.3 Time on Position

The following is the time on position (TOP)³¹ breakdown based on an eight-hour workday and compares the DCA ATCT numbers to that of the two CPCs and TMC that were working in the tower at the time of the accident (the OS and OS-IT are not considered in TOP).³²

Table 2 shows averages for the six months leading up to the day of the accident from July 29, 2024, through January 28, 2025, of the facility, as well as the ALC, LC, and GC that were working on the day of the accident. Table 3 shows the same comparison for the day of the accident only. The TOP column represents the time spent on a control position, the TOP+OD column represents the total time spent on a control position and total time spent performing other required duties, and the % TOP column is the percentage of the workday that was spent working a control position, and the % TOP+OD is the percentage of the workday spent on position or performing officially assigned or required other duties. All time values are shown in hours and minutes (HH+MM).

Table 2. Average time on position and other duties information for all operational positions combined and the LC, ALC, and GC controllers separately.

6-Month (Jul-Jan)	TOP	TOP+OD	% TOP	%TOP+OD
DCA ATCT Average	4+06	5+07	51.24%	63.95%
LC Controller Average	4+12	4+42	52.53%	58.65%
ALC Controller Average	4+34	5+09	57.14%	64.38%
GC Controller Average	4+04	4+53	50.89%	61.04%

Table 3. The time on position and other duties information for the day of the accident, average for all operational positions, and actual for the LC, ALC, and GC controllers separately.

January 29, 2025	TOP	TOP+OD	% TOP	%TOP+OD
DCA ATCT Average	3+52	5+41	48.25%	70.95%
LC Controller Actual	3+20	5+48	41.74%	72.45%
ALC Controller Actual	3+38	6+05	45.41%	76.13%
GC Controller Actual	3+14	5+14	40.50%	65.34%

5.4 ATC Training Program

General feedback received from those interviewed was that the overall training program at DCA ATCT was healthy. When asked to classify the effectiveness of the

³¹ Time on position (TOP) refers to the time actually spent signed onto and performing the duties associated to an actual control position. For this calculation, TOP includes CPC, CPC-IT, DEV, and TMC staffing categories.

³² Facility provided TOP documentation is included in Attachment 12: DCA ATCT TOP Data.

overall training program on a scale of 1 to 5 with 5 being the best, all responded with scores between 3.5 to 4.

Some of those interviewed expressed concerns regarding the impact staffing had, or will continue to have, on certification training specifically. The ability to staff required positions and meet the Air Traffic National Training Initiative (NTI)³³ requirements was often a significant challenge. See section 4.4.3 below for more information regarding the NTI.

An electronic/digital version of the Training and Proficiency Record (Form FAA 3120-1) for each of the involved ATC personnel was requested from the FAA.³⁴ At the time of this writing, the ATC/HP group had received all ALC controller's record. Additional guidance on ATC training was provided in FAA Order JO 3120.4S, "Air Traffic Technical Training."³⁵

5.4.1 Certification Training

The facility had a comprehensive facility training order (*DCA ATCT 3120.4J, "DCA ATCT Facility Training"*)³⁶ that outlined the training track for each individual position. Training generally began with varied stages of classroom training, followed by stages of simulation training and on-the-job training (OJT), all documented through instruction/evaluation reports until reaching recommendation and ultimately certification.

5.4.2 Simulator Training

The DCA ATCT has a fully integrated Tower Simulation System (TSS). The TSS was used to train controllers on airport tower operations. As of May 13, 2025, the FAA indicates that the TSS installed at DCA ATCT has received the latest "Tech Refresh" which updated the TSS software adaptation and performance capabilities.

According to interviews with DCA ATCT personnel, all mentioned that they used the TSS throughout multiple phases of learning, including initial, recurrent, as well as ad-hoc training. At the time of the on-site investigation, according to facility personnel, support from Adacel (the FAA contract provider of the installed TSS) was not consistent, and that they were still trying to work out software issues in the updated system. These issues had resulted in the facility being unable to update or refine any of the previously built scenarios already loaded into the TSS.

³³ See section F, sub-section 5.4.3 below for more information regarding the NTI.

³⁴ Electronic and paper scanned Form FAA 3120-1s are included in Attachment 13: Individual Training and Proficiency Records.

³⁵ A copy of FAA Order JO 3120.4S is located in Attachment 14: FAA Order JO 3120.4S - Air Traffic Technical Training.

³⁶ A copy of DCA ATCT 3120.4J is located in Attachment 15: DCA ATCT 3120.4J - DCA ATCT Facility Training.

5.4.3 Air Traffic National Training Initiative

In July 2019 [updated January 13, 2025], the FAA implemented the NTI³⁷, which was described as a “joint expectation” and not policy. This was jointly developed guidance intended to provide a more efficient and effective progression of trainees toward CPC status. This guidance provided minimum training hours per week for all ATC trainees, broken down by those that were simultaneously required to maintain currency and those that were not as outlined below:

- *If a developmental/CPC-IT is not required to maintain currency, the expectation is 15-18 hours per week.*
- *If a developmental/CPC-IT was required to maintain currency, the expectation is 12-15 hours per week.*

According to the joint expectation guidance, there is a continually updated list of impediments that are to be used to explain when a trainee is unable to achieve the weekly training hour target goal. Two weekly reports are generated documenting the OJT hours accomplished and impediments for shortfalls. The general manager (GM) / assistant regional vice president (ARVP) report was generated by the facility and the director of operations (DO) / regional vice president (RVP) report was generated by the district. The following were the basic content of these reports:

- *The weekly GM/ARVP OJT report will contain the following:*
 - *The OJT hours for each facility and whether or not the training expectation was met/exceeded or not met*
 - *The top impediments to training in the district and each facility*
 - *Access to the weekly facility OJT hours report*
- *The weekly DO/RVP OJT report will contain the following:*
 - *The OJT hours for each district and whether or not the training expectation was met/exceeded or not met*
 - *The top impediments to training in the district and each facility*
 - *Access to the weekly facility OJT hours report*

Both management officials and workforce controllers interviewed expressed concern with the NTI’s impact on trainees and those providing on-the-job training, since its implementation. Those interviewed stated that the NTI was enforced as strict policy and that nationally approved impediments were not accepted by their GM, and that hours missed, regardless of reason, were expected to be made up in the same week. For example, if a person was on vacation one day of their week, the training hours missed for that day were expected to be rolled into the remaining

³⁷ A copy of the most recent FAA/NATCA NTI Joint Expectation Letter is included in Attachment 16: Air Traffic National Training Initiative - January 2025

days, increasing the expected training hours on those days to levels that were not only difficult to achieve, but forced controllers and instructors into a situation they described as “burnout.”

An update to the FAA/NATCA NTI joint expectation letter dated January 13, 2025, stated in part;

The reason the hours are an expectation and not a mandate is because both parties recognize there will be impediments that are for a proper reason and impediments that need to be corrected and or addressed.

It is not the intent of this training initiative to needlessly open positions and/or burn training hours with little to no traffic just to meet the weekly expectations.

Another DCA ATCT union representative provided documentation illustrating the correlation between the amount of training time required to certify prior to NTI implementation, and after. This information indicated that it took less overall time to certify prior to NTI implementation. According to the union representative and others interviewed, this was largely attributed to the facility being allowed to target a trainee’s OJT to be conducted when the most meaningful training for each individual based on their position in the training track. Under the NTI, the daily hours were expected to be met regardless of the situation, often resulting in a training period that was unproductive due to being well above or below their current training level.

At the time of this writing, the ATC/HP group continued to investigate the reasoning behind the reported disparity in interpretation of the NTI guidance, and its impact on training and certification at DCA.

5.4.4 Recurrent Training

According to FAA Order JO 3120.4R, “Air Traffic Technical Training,” recurrent training was defined as:

Recurrent Training is collaboratively developed and delivered national safety training. This training is delivered via electronic means, instructor-led presentations, or any combination thereof. Recurrent Training is intended to increase air traffic controller proficiency, enhance awareness of human factors affecting aviation, and promote behaviors essential for the identification, mitigation, and/or management of risk. Recurrent Training consists of 16 hours of training, conveyed in 8-hour sessions, conducted twice per year. Content need not be duplicated in facility Refresher Training if covered in the Recurrent Training curriculum within the current calendar year. When Recurrent Training is prescribed, AJI must provide instructions for its applicability and specific instructions for documentation in FAA Form 3120-1 or the electronic equivalent.

At the time of this writing, the ATC/HP group continued to review the significant amount of requested training documentation to determine what recurrent training was required, and if/when each of the involved ATC personnel completed it.

5.4.5 Refresher Training

According to FAA Order JO 3120.4S, "Air Traffic Technical Training," refresher training was defined as:

Refresher Training is conducted to maintain and update previously learned knowledge and skills. All operational personnel must complete refresher training. Each facility must develop a written annual Refresher Training plan and conduct the planned training throughout the calendar year. Refresher Training need not be delivered all at once. Refresher Training topics and delivery methods (e.g., electronic learning, classroom, simulation) will be developed by the TA and the Local Safety Council (LSC) or locally developed by another collaborative group. Facilities with access to simulation capabilities must identify which topics will be covered in simulation training from the annual refresher training plan. Simulation training must consist of a cumulative total of not less than two hours per calendar year. Facilities may design their refresher training program specifically for operational areas within the facility that perform different types of operations. The list below is not all-inclusive, and facilities may add other topics in order to meet their needs.

At the time of this writing, the ATC/HP group continued to review the significant amount of training documentation received to determine if/when each of the involved ATC personnel completed it. The DCA ATCT 2024 Annual Refresher Training Plan³⁸ included the following required training including the training delivery method as published:

- | | |
|-----------------------------------|---------------|
| • VFR Aircraft Encountering IMC | TEAM Briefing |
| • PIREPs | TEAM Briefing |
| • Airfield Lighting Control Panel | TEAM Briefing |
| • DCA ATCT SOP | Verbal |
| • Line Up and Wait | TEAM Briefing |
| • Opposite Direction Operations | TEAM Briefing |
| • VIP Movement | TEAM Briefing |
| • Backup Equipment | TEAM Briefing |
| • Lost Aircraft Orientation | TEAM Briefing |
| • Phraseology | TEAM Briefing |
| • Contingency Plan | TEAM Briefing |
| • OS/CIC Training | TEAM Briefing |

³⁸ A copy of the DCA ATCT 2024 refresher training plan is included in Attachment 17: DCA ATCT 2024 Refresher Training Plan.

- | | |
|------------------------------|---------------|
| • Fatigue Awareness | eLMS Assigned |
| • OJTI | TEAM Briefing |
| • Recovery in ATC Operations | Simulation |
| • Safety Alerts | Simulation |

5.4.6 Training Pertinent to Event

There were a number of training entries classified as either supplemental or refresher, that the ATC/HP group considered germane or pertinent to the dynamics related to this event. These training topics have been listed below with a record of attendance by involved personnel in this event.³⁹

June 2023: Maintaining Positive vs Reactive Control - ATSAP⁴⁰ SAFE Discussion

Synopsis: Controllers were reminded that out of the ordinary situations are encountered often. Techniques to be proactive vs reactive were discussed. Topics included *Trust but Verify* and *Traffic for one is Traffic for the other, ask questions if you do not understand what a pilot is requesting*.

- LC No documentation found
- ALC Final training records pending
- GC No documentation found
- CD No documentation found
- OS No documentation found

November 2023: Traffic Advisories - ATSAP SAFE Discussion

Synopsis: Discussed the importance of timely issuance of traffic advisories to pilots. Explained how traffic advisories were used to paint the picture of applicable traffic to pilots.

- LC November 8, 2023
- ALC Final training records pending
- GC November 8, 2023
- CD October 17, 2024
- OS No documentation found (*gap in training documentation Between September 2023 and January 2024*).

³⁹ Copies of the referenced training documents for each of the topics below that could be located are included in Attachment 18: DCA ATCT Pertinent Training Information.

⁴⁰ Air Traffic Safety Action Program (ATSAP) is the FAA's voluntary safety reporting program for ATC personnel based on the Aviation Safety Action Program (ASAP) model. A defined partnership model where safety issues are resolved through corrective action rather than through punishment or discipline. The program includes collection, analysis, and retention of safety data reviewed and acted upon by the FAA and NATCA.

June 2024: Duty Familiarization - Operational Engagement - Position Relief Briefing - Scanning - CTRD Use

Synopsis: The briefing emphasized engagement in the operation to include cultivating a safety culture where employees felt empowered to identify and solve problems. Scanning was reviewed to include scanning priorities and a reminder to stand up to mitigate obstructed views. A review of the CTRD included various ways in which controllers could utilize the equipment to assist to provide information and instruction to aircraft operating within the surface area for which the tower has responsibility.

- LC June 30, 2024
- ALC Final training records pending
- GC June 25, 2024
- CD June 24, 2024
- OS June 28, 2024

September 2024: Mixed Fleet and Helicopter Operations - CISP⁴¹ Discussion

Synopsis: Safety Reports indicated that there was potential for incidents where helicopters and fixed wing aircraft could quickly develop into unexpected and unsafe situations.

- LC September 24, 2024
- ALC Final training records pending
- GC October 10, 2024
- CD September 17, 2024
- OS September 25, 2024

October 2024: Class Bravo Services and Procedures

Synopsis: This training was due to highlighted concerns made by the Air Line Pilots Association (ALPA) regarding airspace excursions. In response, all ATC facilities that provided ATC services in class B airspace were required to be briefed on six items that would provide a refresher on applicable topics in JO 7110.65.

- LC October 10, 2024
- ALC Final training records pending
- GC October 10, 2024
- CD October 17, 2024
- OS October 23, 2024

⁴¹ Confidential Information Share Program (CISP) is the proactive assessment of safety through the sharing and analysis of information collected through the ATSAP and ASAP. These programs support the collection, assessment, and review of data on safety events from the perspectives of air traffic controllers and pilots

January 2025: Recovering from the Unexpected – CISP SAFE Discussion

Synopsis: When unexpected situations occur communication must focus on recovering from the situation using clear phraseology. This briefing was highlighting recovery for TCAS RA⁴² events, however, the lesson in clear communication transcends events.

- LC January 14, 2025
- ALC Final training records pending
- GC January 14, 2025
- CD January 21, 2025
- OS January 29, 2025

5.5 Local Safety Council

The local safety council (LSC)⁴³ appeared highly active in the facility. Specifically, and relative to this event, on September 5, 2024, the committee discussed go-around procedures and explored missed approach points that would avoid headings to help alleviate pilot/controller confusion and workload. The facility actively utilized the LSC to report and elevate complaints. The committee had been responsible for developing a helicopter outreach group. The committee reported that the squadron from which the "PAT" helicopters were assigned, had been invited to these meetings but had not attended.

The LSC was also heavily involved in the quality control processes like developing corrective action plans (CAPs)⁴⁴, conducting system service reviews (SSRs)⁴⁵, systemic issue reviews (SYSIRs)⁴⁶, and quality control (QC) operational skills

⁴² Traffic Collision Avoidance System (TCAS) Resolution Advisory (RA) – A command given to pilots via the TCAS system when it detects a threat of collision. An advisory is provided that includes instructions to climb, descend, or adjust vertical speed in order to increase vertical separation with another aircraft.

⁴³ An LSC is a collaborative team that comprises at least one union representative and one management representative. By mutual agreement, the parties at the local level may designate additional representatives to participate on the LSC. All new LSC members will attend the training provided by the National PFS Program Office prior to gaining access to the PFS Tools. After coordinating with the National PFS Program Office, higher level LSCs (e.g., District, which are still considered local) may form within the mutual agreement of the union and the agency.

⁴⁴ CAPs are collaborative activities enacted to correct non-compliance and areas of risk that have been properly identified, validated, and understood through data collection and analysis. A CAP contains a description of the mitigation actions, the scope of the CAP, a timeframe for completion, a defined monitoring plan, and a defined effectiveness target

⁴⁵ SSRs are conducted to review the air traffic services provided in any situation at any time under any circumstances in an in depth, comprehensive, and collaborative manner. The absence of a defined list of instances when an SSR must be conducted allows a SDP the latitude to use judgment and discretion in the determination of when to perform an SSR. SSRs may be specifically focused and limited in scope or may eventually evolve into a larger more in-depth review.

⁴⁶ SYSIRs are a comprehensive and collaborative review of potential systemic issues identified during OSAs, SSRs, CSBR, and CERs. SYSIRs are a method for SDP personnel that conduct an OSA, SSR, CSBR, or CER to identify areas that have potential system impact, whether at the SDP, service area, or service

assessments (OSAs)⁴⁷. They communicated regularly with the event review committee (ERC)⁴⁸ regarding local ATSAP issues and collaboratively worked to mitigate them. The overall health of the DCA ATCT LSC appeared excellent. All personnel interviewed expressed high regard for their LSC and the work they did.

6.0 Safety Management System

Requirements and guidance regarding safety management systems (SMS) and safety risk management (SRM) were documented for FAA air traffic organization (ATO) across multiple publications.

FAA Order 1100.161B "Air Traffic Safety Oversight"⁴⁹, chapter 3, section 4 "Safety Risk Management", stated, in part;

The ATO's SMS must develop, maintain, and use a formal, documented methodology to provide critical data and information for decision makers by identifying hazards, analyzing safety risk, assessing safety risk, and mitigating safety risk to an acceptable level.

a. ATO must use SRM for implementing new systems, revising existing systems, developing or revising operational procedures, and for hazards or ineffective risk controls identified through Safety Assurance processes.

(1) When conducting SRM, ATO must determine if hazards with acceptable risk (i.e., medium- and low-risk) can be further mitigated.

(2) All identified hazards impacting ATO operations must be tracked.

b. ATO must conduct the SRM process at the appropriate level of complexity and rigor for the potential hazard's scope and immediacy. New systems; changes to existing systems; and new or revised operational procedures, including those at the local level, can only be implemented after the SRM process indicates the change will meet or exceed the defined acceptable

unit level. SYSIRs can also be generated directly through the Air Traffic SYSIR function in CEDAR if a suspected systemic issue is found outside of the OSA, SSR, CSBR, or CER process. Individuals who identify potential systemic issues initiate the process of identifying previously unrecognized risk. When conducting an OSA, SSR, or CER, SDP personnel must continually look for underlying issues beyond the initial "what" that may have occurred and look for the "why." Asking "why" will help lead personnel to the initial identification of a systemic issue.

⁴⁷ A QC OSA is a QC sampling method used to collect data by assessing the technical performance demonstrated by operational personnel. A QC OSA must not be used as the basis of, or to support, IPM.

⁴⁸ A three-member group comprising representatives from each party to the respective VSRP non-punitive safety-reporting program. The ERC reviews and analyzes submitted confidential reports to determine acceptability of the report under the VSRP requirements, to identify actual or potential safety problems of accepted reports, and to ensure the appropriate follow-up action is taken and implemented for resolution when appropriate.

⁴⁹ A copy of this order is included in Attachment 19: FAA Order 1100.161B - Air Traffic Safety Oversight.

safety level. During the SRM process, ATO must examine and analyze all related hazards, including existing hazards. The existing hazards must be examined to validate that their risk level has not increased and to determine if additional mitigation is warranted.

FAA Order JO 1000.37C "Air Traffic Organization Safety Management System"⁵⁰, chapter 1, section 1 "Purpose of this order", stated, in part:

This order establishes Safety Management System (SMS) policies for the Air Traffic Organization (ATO). It defines the scope, requirements, and applications of the SMS in the ATO and gives the responsibility for owning and executing the SMS to all employees at all levels of the ATO, from the ATO Chief Operating Officer (COO) to the individual air traffic controllers and airway transportation systems specialists at a Service Delivery Point (SDP). The ATO COO is the ultimate accountable executive for ensuring the effectiveness of the SMS.

FAA Order JO 7200.20B "Voluntary Safety Reporting Programs"⁵¹ outlines the Voluntary Safety Reporting Programs (VSRPs) implemented by ATO. The order stated, in part:

VSRPs are a key component of the ATO Safety Management System (SMS), providing a method to identify and correct potential safety hazards. These important programs encourage employees involved in the delivery of air traffic services to submit safety reports voluntarily and are foundational to a healthy safety culture.

Chapter 2, section 11 "What may be reported to a VSRP?" of the same manual stated, in part:

VSRPs are for reporting non-compliance with FAA directives and aviation safety concerns. Regardless of position or whether any additional investigation takes place, an eligible employee may always file a VSRP report when they observe a safety problem or experience a safety-related event.

FAA Order 8040.4C "Safety Risk Management Policy"⁵², chapter 1, section 8 "Scope", stated in part:

It describes the principles used to guide SRM within the FAA. It formalizes the use of SRM across the FAA, describes the specific steps to use when performing

⁵⁰ A copy of this order is included in Attachment 20: FAA Order JO 1000.37C - ATO Safety Management System.

⁵¹ A copy of this order is included in Attachment 21: FAA Order JO 7200.20B - Voluntary Safety Reporting Program.

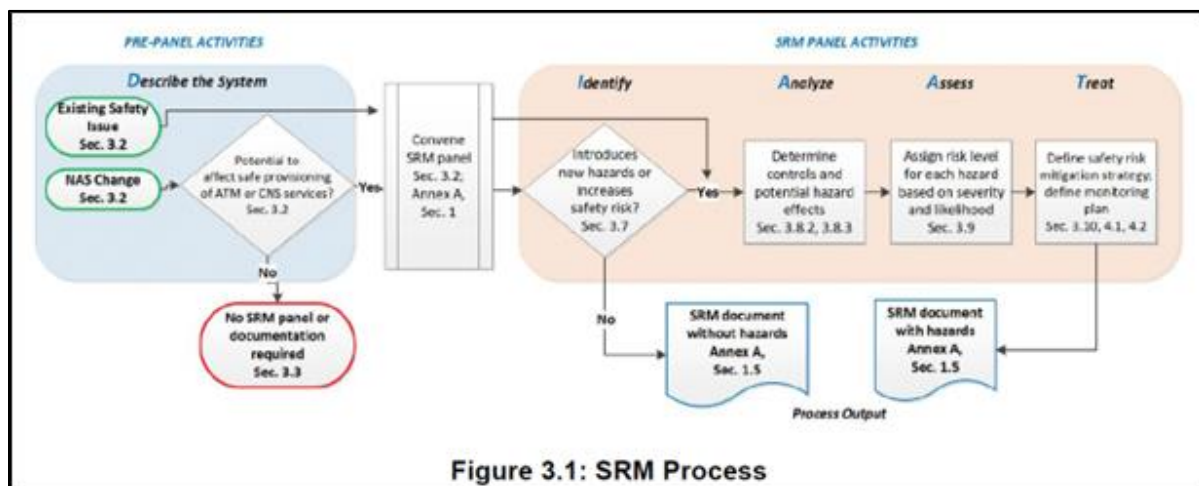
⁵² A copy of this order is included in Attachment 22: FAA Order 8040.4C - Safety Risk Management Policy.

SRM, and enables communication and coordination across FAA organizations for enhanced safety risk decision making.

According to the DCA ATCT Operations Manager, the facility utilized ATSAP and the LSC to address safety concerns at the facility. If changes to the LOA were needed, the facility would go through a local SRM process; a more formal process was used with outside facilitators, for example, when making changes to the helicopter chart. According to JO 7200.20B, an ATSAP should be submitted “when an individual is involved in, observes, or identifies an operational safety hazard/problem, or experiences a safety-related event.”

The ATO SMS Manual⁵³, Section 3 The Safety Risk Management (SRM) Process, defined the scope of the SRM process, when to perform SRM, SRM process phases, and how to use SRM to support changes in the NAS. Subsection 3.1 Scope of the Safety Risk Management Process, stated, in part:

The Air Traffic Organization (ATO) Safety Risk Management (SRM) process is used to determine the safety risk of National Airspace System (NAS) changes or existing safety issues associated with the provision of air traffic management services.” It also stated, “This section provides a linear SRM process to follow, guidelines to identify safety hazards and mitigate their risks, and requirements for the development of consistent and thorough safety analyses. Using the steps in this section to perform SRM will not always result in an exhaustive study of air traffic procedures, operations, or NAS equipment (i.e., hardware and software). The appropriate level of detail used when conducting SRM depends on the complexity, size, and potential effect of the NAS change or existing safety issue. Figure 3.1 provides a high-level depiction of the key steps, decision points, and outputs of the SRM process.



⁵³ A copy of this manual is included in Attachment 23: FAA ATO Safety Management System Manual.

The ATO SMS Manual, Section 3 The Safety Risk Management (SRM) Process, Subsection 3.2 When to Perform SRM, stated, in part:

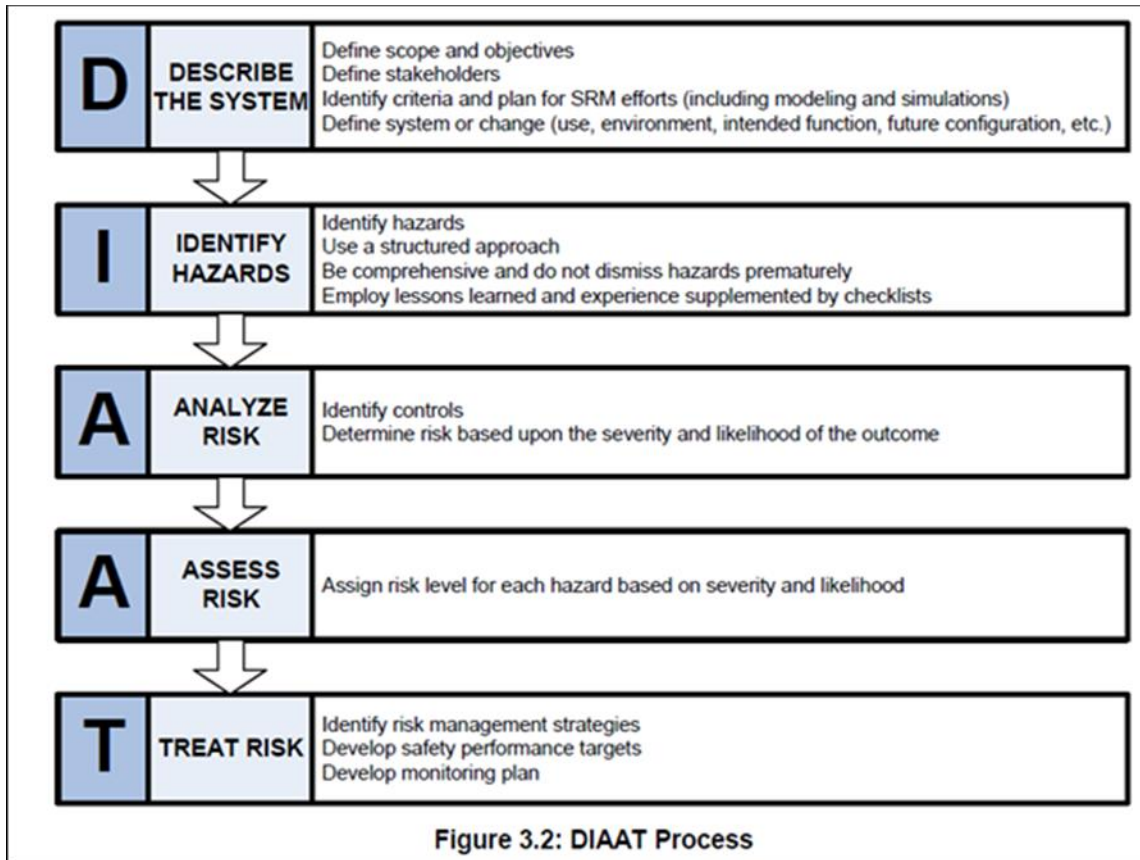
SRM is most frequently performed in response to a NAS change. NAS changes may be proposed and initiated as part of implementation plans for new/modified air traffic procedures, operations, or NAS equipment, or in response to existing safety issues currently in the NAS. For the ATO, a NAS change is a modification to any element of the NAS that pertains to, or could affect, the provision of air traffic management and/or communication, navigation, and surveillance services. Air traffic controllers and technicians, their training, and their certification are elements of the NAS and directly relate to the provision of air traffic services.

*In some cases, SRM is performed in response to a request to take action on an existing safety issue. **Existing safety issues** are existing contributing factors or findings that led to, or could lead to, an unsafe outcome. Requests for action to address such issues may be proposed and initiated as part of a Safety Assurance function. This is usually a result of Quality Assurance, audits, or assessments. If a request to take action on an existing safety issue is received, SRM must be performed.*

Though not all NAS changes will require SRM, the decision and justification to forgo performing SRM is a safety decision.

The ATO SMS Manual, Section 3 The Safety Risk Management (SRM) Process, Subsection 3.5 SRM Process Phases, stated, in part:

SRM is composed of a five-phase process called the DIAAT, presented in Figure 3.2.



G. INTERVIEWED PERSONNEL

Interviews were conducted with all involved (working in the tower at the time of the accident) DCA ATCT personnel. Additionally, current and former ATC personnel from DCA ATCT and PCT TRACON, and the Washington district TMO were interviewed, none of whom were present or working in the tower at the time of the accident. All interviews were recorded and transcribed by a court reporting and transcription service. Transcripts were reviewed by the ATC/HP group, as well as those interviewed, and corrections were recertified and required clarifications shown through footnotes.⁵⁴

NOTE: The individual training and proficiency records provided by the FAA had multiple contradictions in both assignment dates, as well as certification dates for position qualifications. Dates provided below are as accurate as possible based on available data. All individual training and proficiency records provided by the FAA are included in attachment 13 of this report, which reflect these contradictions in documentation. Multiple attempts were made to clarify with FAA, and while some were ultimately clarified, some were not at the time of this writing and therefore the dates may contain inaccuracies. The ATC/HP group continue to investigate.

⁵⁴ All interview transcripts are included in Attachment 24: Interview Transcripts

1.0 Involved DCA ATCT Personnel

For the purpose of this investigation, the five personnel working in the tower at the time of the accident are referred to as involved personnel. Of the involved personnel, only the LC, ALC, and OS had any ATC services interaction with, or in support of, the accident aircraft.⁵⁵

1.1 Local Control Controller

The LC controller that was working in the tower at the time of the accident was working the LC and Helicopter Control (HC) positions combined to the LC north position. The LC controller was in direct communication and providing services to both accident aircraft. The LC controller's working hours for the week leading up to and including the day of the accident were:

Table 4. Hours worked by the LC controller for the week leading up to and day of the accident.

Day	Date	Shift	Hours	Total Working Hours
Thursday	Jan 23	Eve	1300-2100	08 hours + 00 minutes
Friday	Jan 24	Day	0700-1500	08 hours + 00 minutes
Saturday	Jan 25	Day	0600-1400	08 hours + 00 minutes
Sunday	Jan 26	Day	0600-1400	08 hours + 00 minutes
Monday	Jan 27	Off	None	None
Tuesday	Jan 28	Off	None	None
Wednesday	Jan 29	Eve	1400-0000	10 hours + 00 minutes

1.1.1 Position Duties and Responsibilities

According to the DCA ATCT SOP, the primary duties and responsibilities of the LC position included (but were not limited to):

- *The LC controller must:*
 - *Provide initial separation between successive departures.*
 - *Provide separation between arrivals and departures.*
- *Coordinate with the final controller if the tower issues any control instruction that affects the sequence or separation established by the final controller.*
- *When it becomes apparent that a loss of standard separation may occur within the tower's airspace, LC must initiate corrective action and coordinate with the appropriate sectors.*
- *Issue ATC instructions that will keep unauthorized aircraft clear of P-56A/B.*

⁵⁵ All documentation on "hours worked" provided by the FAA are included in Attachment 25: Involved Controllers - Hours Worked Documentation.

According to the DCA ATCT SOP, the primary duties and responsibilities of the HC position included (but were not limited to):

- *The HC controller must:*
 - *Separate VFR traffic from DCA arrivals and departures.*
 - *Issue safety alerts and traffic advisories as required.*
- *Clear VFR aircraft on routes or into zones as depicted on the Baltimore-Washington Helicopter Route Chart.*
- *Advise OS of aircraft with invalid mode C.*
- *Issue ATC instructions that will keep unauthorized aircraft clear of P-56.*
- *Authorize a frequency change when the pilot reports or you observe the aircraft approaching the outer limits of DCA airspace.*
- *Ensure aircraft are squawking assigned beacon codes.*
- *Validate mode C of all aircraft operating within the DCA class B.*
- *Radar identify all aircraft requesting class B clearance.*
- *Advise pilots of their radar identification status while operating in the class B.*

1.1.2 Control Position Assignments - Accident Day

The LC controller was scheduled for an eve shift from 1400-2200 on the day of the accident. According to the position logs provided by the FAA⁵⁶, the following were the times that the LC controller was signed onto a control position.

1413-1452	ALC	39 minutes
1545-1659	CD/FD	74 minutes
1748-1835	ALC	47 minutes
1928-2054	LC	86 minutes

1.1.3 Duty Location Assignment History

The LC controller entered on duty with the FAA on June 22, 2016, and began with initial training at the FAA Academy in Oklahoma City, OK. Below is a list of the LC controller's duty assignments in order according to their report date at each.

FAA Academy	June 22, 2016	Student
PIE ATCT	September 20, 2016	Developmental
ILG ATCT	May 12, 2019	CPC
DCA ATCT	October 23, 2022	CPC

⁵⁶ Copies of all logs included in Attachment 26: DCA ATCT Facility Logs and Records.

1.1.4 Position Qualification and Certification

The LC controller was certified on all control positions at DCA ATCT. Below is a list of the LC controller's position certifications according to the certification date of each.

CD, FD	December 30, 2022
GC	August 06, 2023
ALC, LC, HC	May 22, 2024

1.1.5 Recent Activities

The LC controller's recent activities were documented in interviews, cellular telephone records and schedules. The LC controller thought they only needed about 5 hours of sleep to feel rested. They did not nap during the day and routinely consumed caffeine. In general, they usually fell asleep quickly after getting into bed and would wake up early. They did not have any trouble falling asleep or staying asleep and had never talked to a doctor about any sleep issues.

On Sunday, January 26, 2025, the LC controller thought they woke up about 0430 or 0500; they did not set an alarm. The LC controller worked from 0600 to 1400. There was cellular telephone activity between 0645 and 1308. After work they went out for the evening; it was their "Friday". They did not recall what time they went to bed but thought it might have been between 2300-0000.

On Monday, January 27, it was the LC controller's day off and they thought they would have awoken about the same time, 0400-0500 and went to the gym and did routine activities. Cellular activity began at 0746. They thought they might have gone to bed between 2200-0000. Cellular telephone activity ended at 0114 on January 28.

On Tuesday, January 28, the LC controller thought their activities were similar to Monday. It was their day off and they awoke about 0430-0500 and did routine activities. Cellular activity began at 0814. They spent time with a friend who was in town and thought they went to bed about 2230-2300. Cellular telephone activity ended at 2322.

On Wednesday, January 29, the day of the event, The LC controller thought they woke up about 0500. Cellular activity began at 0658. They spent time with their out-of-town friend and then went to work; their shift started at 1400. They had a light meal on shift as they were planning to meet a friend after getting off work. The last cellular activity prior to the accident ended at 1920. They signed onto the LC position at 1928. While on shift the LC controller recalled "feeling fine".

In the 72 hours prior to the event, the LC controller did not recall taking any prescription or nonprescription medication that might have affected their performance.

1.1.6 Medical Clearance

The LC controller held a valid medical clearance on the day of the accident which was sighted by the group chairman while on-site. The LC controller's most recent aviation physical examination was conducted on August 15, 2024. For additional medical clearance information see the Medical Factual Group factual report in the docket.

1.2 Assistant Local Control Controller

The ALC controller that was working in the tower at the time of the accident was working ALC standalone from the ALC north position. The ALC controller's working hours for the week leading up to and including the day of the accident were:

Table 5. Hours worked by the ALC controller for the week leading up to and day of the accident.

Day	Date	Shift	Hours	Total Working Hours
Thursday	Jan 23	Off	None	None
Friday	Jan 24	Off	None	None
Saturday	Jan 25	Off	None	None
Sunday	Jan 26	Off	None	None
Monday	Jan 27	Off	None	None
Tuesday	Jan 28	Off	None	None
Wednesday	Jan 29	Eve	1400-0000	10 hours + 00 minutes

1.2.1 Position Duties and Responsibilities

According to the DCA ATCT SOP, the primary duties and responsibilities of the ALC position included (but were not limited to):

- Alert LC of any unusual situations or traffic conflicts.
- Maintain surveillance of the local traffic pattern and landing area.
- Assist LC with monitoring aircraft on final via CTRD.
- Assist LC with sequencing departures to comply with release times, clearance void times, and any other restrictions.
- Review proposed departure times to ensure that flight plans do not time out.
- Unless otherwise requested by LC, place the sequenced departure strips in the LC strip bay, in departure order, from bottom to top.
- Unless otherwise requested by LC, as aircraft depart, scan departure strips and offset them in the departure bay. If the EFSTS is unusable, call the

appropriate departure sector with a boundary check and offset the strip in the departure bay.

1.2.2 Control Position Assignments - Accident Day

The ALC controller was scheduled for an eve shift from 1400-2200 on the day of the accident. According to the position logs provided by the FAA, the following are the times that the ALC controller was signed onto a control position or TMC/CIC.

1415-1429	TMC	14 minutes
1446-1635	GC	109 minutes
1733-1822	CIC/TMC	49 minutes
1823-1824	TMC	1 minute
1825-1859	GC	34 minutes
1956-2054	ALC	58 minutes

1.2.1 Duty Location Assignment History

The ALC controller entered on duty with the FAA on January 19, 2011, and began with initial training at the FAA Academy in Oklahoma City, OK. Below is a list of the ALC controller's duty assignments in order according to their report date at each.

FAA Academy	January 19, 2011	Student
BTR ATCT	April 15, 2011	Developmental
ILG ATCT	September 8, 2013	Developmental, CPC, CIC
IAD ATCT	October 14, 2018	CPC-IT, CPC, CIC
DCA ATCT	January 2, 2022	CPC-IT, CPC, TMC

1.2.2 Position Qualification and Certification

The ALC controller was certified on all control positions at DCA ATCT, as well as TMC. Below is a list of the ALC controller's position and TMC certifications according to the certification date of each.

CD, FD	March 3, 2022
GC	July 17, 2022
ALC, LC	February 19, 2023
HC	February 20, 2023
TMC	March 27, 2023

1.2.3 Recent Activities

The ALC controller's recent activities were documented in interviews, cellular telephone records and schedules. They did not nap in the 72 hours preceding the accident. They occasionally drank caffeinated beverages. They did not have any

problems falling asleep or staying asleep at night and had never talked to a doctor about any sleep issues.

On Sunday, January 26, 2025, the ALC controller was on vacation. They thought they woke up about 0900-0930. They spent the day at the beach. They thought they went to bed about 2200. They did not watch TV or read before bed and fell asleep quickly. There was no cellular telephone activity.

On Monday, January 27, the ALC controller thought they woke up about 0700 and returned home from vacation. They thought they went to bed about 2030-2100. Cellular telephone activity began at 0648 and was intermittent throughout the day; activity ended at 2043.

On Tuesday, January 28, the ALC controller thought they woke up about 0400 and went on shift at 0600. They left work at noon for an appointment. They had dinner with a friend and went to bed about 2200. There was cellular activity between 0819 and 0014 on January 29.

On Wednesday, January 29, the ALC controller thought they woke up about 0830 with an alarm and took their son to an appointment and to camp. Cellular activity began at 0954. Their shift started at 1400. They reported feeling "fine" on shift. Cellular telephone activity ended prior to the accident at 1943. The ALC controller signed onto the ALC position at 1956.

In the 72 hours prior to the event, the ALC controller did not recall taking any prescription or nonprescription medication that might have affected their performance.

1.2.4 Medical Clearance

The ALC controller held a valid medical clearance on the day of the accident which was sighted by the group chairman while on-site. The ALC controller's most recent aviation physical examination was conducted on April 11, 2024. For additional medical clearance information see the Medical Factual Group factual report in the docket.

1.3 Ground Control Controller

The GC controller that was working in the tower at the time of the accident was working GC standalone from the GC north position. The GC controller's working hours for the week leading up to and including the day of the accident were:

Table 6. Hours worked by the GC controller on the week leading up to and day of the accident.

Day	Date	Shift	Hours	Total Working Hours
Thursday	Jan 23	Off	None	None
Friday	Jan 24	Off	None	None
Saturday	Jan 25	Mid	2100-0700	10 hours + 00 minutes
Sunday	Jan 25-26	Mid	2100-0700	10 hours + 00 minutes
Monday	Jan 27	Off	None	None
Tuesday	Jan 28	Eve	1345-2200	08 hours + 15 minutes
Wednesday	Jan 29	Eve	1300-0000	11 hours + 00 minutes

1.3.1 Position Duties and Responsibilities

According to the DCA ATCT SOP, the primary duties and responsibilities of the GC position included (but were not limited to):

- *Control aircraft and vehicles operating on movement areas, excluding runways.*
- *When airport resource management tool (ARMT) is out of service, verbally request departure releases from the TMC, FD, or OS.*
- *Review proposed departure times and ensure that flight plans do not time out.*
- *Notify the OS/CIC of opposite direction operations (ODO) requests.*

1.3.2 Control Position Assignments - Accident Day

The GC controller was scheduled for an eve shift from 1400-2200 on the day of the accident. According to the position logs provided by the FAA, the following are the times that the GC controller was signed onto a control position or CIC.

1315-1359	ALC	44 minutes
1400-1403	CIC	3 minutes
1451-1544	CD/FD	53 minutes
1636-1723	GC	47 minutes
1817-1919	CD/FD	62 minutes
2015-2129	GC	74 minutes

1.3.3 Duty Location Assignment History

The GC controller entered on duty with the FAA on December 15, 2008, and began with initial training at the FAA Academy in Oklahoma City, OK. Below is a list of the GC controller's duty assignments in order according to their report date at each.

FAA Academy	December 15, 2008	Student
SUS ATCT	March 15, 2009	Developmental, CPC
DCA ATCT	May 23, 2011	CPC-IT, CPC, CIC

1.3.4 Position Qualification and Certification

The GC controller was certified on all control positions at DCA ATCT. Below is a list of the GC controller's position certifications according to the certification date of each.

CD, FD	July 1, 2011
GC	October 19, 2011
HC	July 3, 2012
ALC, LC	July 10, 2012
CIC	October 01, 2013

1.3.5 Recent Activities

The GC controller's recent activities were documented in interviews, cellular telephone records and schedules. In general, they did not have any problems falling asleep or staying asleep at night. They never talked to a doctor about any sleep issues. They usually had a cup of coffee in the morning.

On Sunday, January 26, 2025, the GC controller had worked the midnight shift from Saturday night into Sunday morning. They got about 3 hours sleep and woke up about 1100. They stayed home the rest of the day. Cellular telephone activity was between 1747 and 1913. They went to bed at 0130.

On Monday, January 27, the GC controller awoke about 1000. It was their day off and they stayed home. Cellular telephone activity was between 1024 and 1931. They went to bed about 0130.

On Tuesday, January 28, the GC controller awoke about 1100. They were scheduled to work 1200-2200 but took leave and arrived about 1345. Cellular telephone activity was recorded between 1249 and 2126. They went to bed about midnight. There was a cellular telephone call at 0307 for 3 minutes and 29 seconds.

On Wednesday, January 29, the GC controller awoke about 0800. They were scheduled to work 1200-2200 but took an hour of leave and arrived about 1300. They had a meal while on shift. It was a normal shift, and they felt "fine". The GC controller signed onto the GC position at 2015. Cellular telephone activity was recorded between 0836 and 1850.

In the 72 hours prior to the accident, the GC controller did not recall taking any prescription or non-prescription medication that might have affected their performance.

1.3.6 Medical Clearance

The GC controller held a valid medical clearance on the day of the accident which was sighted by the group chairman while on-site. The GC controller's most recent aviation physical examination was conducted on September 3, 2024. For additional medical clearance information see the Medical Factual Group factual report in the docket.

1.4 Clearance Delivery Controller

The CD controller that was working in the tower at the time of the accident was working CD and FD combined from the CD/FD position. The CD controller's working hours for the week leading up to and including the day of the accident were:

Table 7. Hours worked by the CD controller on the week leading up to and day of the accident.

Day	Date	Shift	Hours	Total Working Hours
Thursday	Jan 23	Off	None	None
Friday	Jan 24	Day	0700-1600	10 hours + 00 minutes
Saturday	Jan 25	Day	0600-1400	08 hours + 00 minutes
Sunday	Jan 26	Off	None	None
Monday	Jan 27	Off	None	None
Tuesday	Jan 28	Eve	1300-2200	09 hours + 00 minutes
Wednesday	Jan 29	Eve	1300-0000	11 hours + 00 minutes

1.4.1 Position Duties and Responsibilities

According to the DCA ATCT SOP, the primary duties and responsibilities of the CD position included (but were not limited to):

- *CD normally operates from position 009.*
- *The primary frequency is 128.25.*
- *Review proposed flight plans for anomalies.*
- *Issue ATC clearances, traffic management initiatives, expect departure clearance time (EDCT) times and any other pertinent information as necessary.*
- *Mark strips in accordance with DCA ATCT SOP.*
- *When duplicate flight plans are received, verify the requested flight plan with the pilot and remove the undesired flight plan from the FDIO.*

According to the DCA ATCT SOP, the primary duties and responsibilities of the FD position included (but were not limited to):

- *Remove and post strips from the flight strip printer or copy and post ATC clearances as received from the ARTCC. Replace printer strips as necessary.*

Ensure that all positions of operation in the tower have a supply of blank strips.

- Transmit and receive Flight Data Input/Output (FDIO) messages. Take prompt action to make proper distribution and notification on all revisions and remove strip (RS) messages. Forward general information (GI) messages as appropriate.*
- Forward all traffic management messages to the TMC/OS.*
- When FD is combined with CD, and the TMC position is not staffed, the OS/CIC will coordinate aircraft release times through the ZDC ARTCC TMC or the PCT TRACON TMC, in accordance with the APREQ schedule posted on the IDS.*
- When directed by the OS/CIC, call ZDC ARTCC FD to coordinate reentry of flight plans that have timed out into the FDIO. Forward the new transponder code to the appropriate position.*

1.4.2 Control Position Assignments - Accident Day

The CD controller was scheduled for an eve shift from 1300-2300 on the day of the accident. According to the position logs provided by the FAA, the following are the times that the CD controller was signed onto a control position.

1406-1524	LC	78 minutes	(in training)
1613-1732	ALC	79 minutes	(in training)
1822-1927	LC	65 minutes	(in training)
2021-2111	CD	50 minutes	

1.4.3 Duty Location Assignment History

The CD controller entered on duty with the FAA on August 4, 2019, and was hired under Veterans' Recruitment Appointment (VRA) having previously served as an air traffic controller in the US Navy. Below is a list of the CD controller's duty assignments in order according to their report date at each.

MFD ATCT	August 4, 2019	Developmental, CPC
ADW ATCT	March 27, 2022	CPC-IT, CPC, OS-IT, OS
DCA ATCT	May 23, 2024	OS-IT

1.4.4 Position Qualification and Certification

The CD controller was an OS-IT and was certified on CD, FD, and GC only. Below is a list of the CD controller's position certifications according to the certification date of each.

CD, FD	February 14, 2024
GC	August 1, 2024

1.4.5 Recent Activities

The CD controller's recent activities were documented in interviews, cellular telephone records and schedules. They did not have any problems falling asleep or staying asleep at night. They wore a CPAP⁵⁷ machine while sleeping at night but did not wear it while napping. They had a medical waiver and submitted a CPAP usage report annually, the most recent time prior to the event being in June of 2024. They would drink a cup of coffee throughout the day.

On Sunday, January 26, 2025, The CD controller thought they awoke about 0800. It was a day off. Cellular telephone activity was between 0735 and 1123. They thought they went to bed about 2200.

On Monday, January 27, the CD controller awoke about 0400. They did routine family activities and had some appointments. Cellular telephone activity was between 1131 and 1803. They thought they went to bed about 2200.

On Tuesday, January 28, the CD controller awoke about 0400. Cellular telephone activity was between 0826 and 2153. They worked from 1300-2200. They thought they took a nap that day, but the time was unknown. The time they went to bed was unknown.

On Wednesday, January 29, the CD controller awoke about 0400. They thought they took a nap between 1000 and 1100 then left the house about 1145 for work. They started their shift at 1300. They had a meal while on shift. They recalled feeling "fine" when signing onto the CD position at 2021. Cellular telephone activity was between 1121 and 1759.

In the 72 hours prior to the accident, the CD controller did not recall taking any prescription or nonprescription medication that might have affected their performance.

1.4.6 Medical Clearance

The CD controller held a valid medical clearance on the day of the accident which was sighted by the group chairman while on-site. The CD controller's most recent aviation physical examination was conducted on June 7, 2024. For additional medical clearance information see the Medical Factual Group factual report in the docket.

⁵⁷ Continuous positive airway pressure (CPAP) is a machine commonly used to treat sleep apnea by delivering continuous air through the mouth and/or nose to help keep a person's airways open while sleeping. For more information, see: [CPAP Machine: What It Is, How It Works & Side Effects](#)

1.5 Operations Supervisor

The OS that was working in the tower at the time of the accident was working OS standalone from the OS position. The OS's working hours for the week leading up to and including the day of the accident were:

Table 8. Hours worked by the OS on the week leading up to and day of the accident.

Day	Date	Shift	Hours	Total Working Hours
Thursday	Jan 23	Eve	1245-2200	09 hours + 15 minutes
Friday	Jan 24	Eve	1200-2100	09 hours + 00 minutes
Saturday	Jan 25	Day	0830-1500	06 hours + 30 minutes
Sunday	Jan 25-26	Off	None	None
Monday	Jan 27	Off	None	None
Tuesday	Jan 28	Off	None	None
Wednesday	Jan 29	Eve	1300-0000	11 hours + 00 minutes

1.5.1 Position Duties and Responsibilities

According to the DCA ATCT SOP, the primary duties and responsibilities of the OS position included (but were not limited to):

- *The OS normally operates from position 008.*
- *Provide operational supervision and direct the tower operation to ensure an efficient flow of air traffic.*
- *Make necessary notifications of:*
 - *Suspected operational errors.*
 - *Pilot deviations.*
 - *Near midair collision (NMAC) reports*
 - *Other incidents requiring notifications.*
- *Combine and de-combine positions.*

According to the DCA ATCT SOP, the primary duties and responsibilities of the OS when required to perform TMU duties included (but were not limited to):

- *Participate in the development of daily traffic management strategies.*
 - *Determine final separation interval.*
 - *Ensure arrival rate is consistent with predicted airport capacity.*
- *Inform the PCT TRACON Mount Vernon OS/CIC of conditions that may require traffic management initiatives.*
- *Advise the PCT TRACON MTV OS/CIC when requested final interval is not being delivered.*
- *Assume TMC duties when the DCA ATCT TMC is not staffed.*

1.5.2 Control Position Assignments - Accident Day

The OS was scheduled for an eve shift from 1300-2300 on the day of the accident. According to the position logs provided by the FAA, the following are the times that the OS was signed onto a control position or OS.

1750-1821	LC	31 minutes
1630-1732	OS	62 minutes
1733-1747	ALC	14 minutes
1823-2129	OS	186 minutes
2200-2205	OS	5 minutes

1.5.3 Duty Location Assignment History

The OS entered on duty with the FAA on July 19, 2011, and began with initial training at the FAA Academy in Oklahoma City, OK. Below is a list of the OS's duty assignments in order according to the report date at each.

FAA Academy	July 19, 2011	Student
FPR ATCT	September 12, 2011	Developmental, CPC
DCA ATCT	April 19, 2015	CPC-IT, CPC
MIA ATCT	June 11, 2017	CPC-IT
DCA ATCT	June 10, 2018	CPC-IT, CPC, CIC, OS

1.5.4 Position Qualification and Certification

The OS was reportedly certified on all control positions at DCA ATCT. Below is a list of the OS's position certifications in order according to the certification date of each. These are re-certification dates because they had been previously certified on all but OS when they were assigned to DCA ATCT from 2015-2017.

FD	June 22, 2018
CD	June 23, 2018
GC	July 3, 2018
ALC, HC, LC	Documentation not found for re-certifications.
OS	May 23, 2024

1.5.5 Recent Activities

The OS's recent activities were documented in interviews, cellular telephone records and schedules. In general, when not working, they would wake up about 1000, except for Saturday mornings when they had to work. They would usually go to bed about 0100-0200. They would fall asleep with the TV on, and they or their partner would turn it off during the night; they said it would depend but they would usually fall back to sleep quickly after that. They did not have any problems falling

asleep or staying asleep and had never talked to a doctor about any sleep issues. They did not have any recent injury or illness.

On Sunday, January 26, 2025, the OS was not working. They did not recall what time they woke up and did not wake up to an alarm. They were off shift and spent the day watching football. They did not recall what time they went to bed. There was cellular telephone activity at 1923, from 0044 until 0122 and at 0259.

On Monday, January 27, the OS thought they woke up about 1000. They did not work on Monday and stayed home and then hung out with friends and watched the basketball game. The OS did not recall when they went to bed. Cellular telephone activity began at 0834 and was intermittent throughout the day; it ended at 2052.

On Tuesday, January 28, the OS thought they woke up about 1000 and did not work. They were at home all day but did not recall their specific activities. Cellular telephone activity was recorded at 0954 and 1943. They thought they went to bed about 0100-0200.

On Wednesday, January 29, the OS thought they woke up about 1000 and arrived at work at 1300. They had a meeting and checked emails, then went up to the tower cab about 1625. They ate one meal before going up to the tower cab and one meal and a yogurt while in the tower cab. The OS said they drank a lot of caffeine, about 8 cups a day, and did so that day. They felt good that day and signed onto the OS position at 1823. Cellular telephone activity began at 1024 and ended before the accident at 1753.

In the 72 hours prior to the event, the OS did not recall taking any prescription or nonprescription medications that might have affected their performance.

1.5.6 Medical Clearance

The OS held a valid medical clearance on the day of the accident which was sighted by the group chairman while on-site. The OS's most recent aviation physical examination was conducted on February 8, 2024. For additional medical clearance information see the Medical Factual Group factual report in the docket.

2.0 Non-Involved Personnel

2.1 DCA ATCT Operations Manager

The OM had been assigned to DCA ATCT since February 2013 but had only been serving as OM since March 2023. They had previously served as a CPC and OS before being selected as OM.

2.1.1 Duty Location Assignment History

The OM entered on duty with the FAA on August 29, 2008, and began with initial training at the FAA Academy in Oklahoma City, OK. Below is a list of the OM's duty assignments in order according to their report date at each.

FAA Academy	August 29, 2008	Student
PNE ATCT	November 17, 2008	Developmental, CPC
DCA ATCT	February 10, 2013	CPC-IT, CPC, OS-IT, OS, OM

2.2 DCA ATCT Air Traffic Manager

The ATM had been assigned as ATM since June 30, 2024, but was only acting and was officially still assigned as OM at PCT TRACON where they were serving as XO (acting) before being assigned to DCA ATCT. They had served as a CPC at DCA ATCT earlier in their career from 2011-2014.

2.2.1 Duty Location Assignment History

The ATM entered on duty with the FAA on November 19, 2010, and began with initial training at the FAA Academy in Oklahoma City, OK. Below is a list of the ATM's duty assignments in order according to their report date at each.

FAA Academy	November 19, 2010	Student
DCA ATCT	February 20, 2011	Developmental, CPC
ZDC ARTCC	October 5, 2014	CPC-IT, CPC
AOV-140	January 22, 2017	SMS / Future Systems / Branch Manager
ATSCC-DEN	September 1, 2019	Air Traffic Security Coordinator
PCT TRACON	January 16, 2022	OM, XO-A
DCA ATCT	June 30, 2024	ATM

2.3 PCT TRACON Air Traffic Manager

The PCT TRACON ATM had been assigned as ATM at PCT TRACON since February 6, 2022, and had been serving in some capacity at PCT TRACON since 2009. The ATM had an extensive knowledge of operations in the DC area.

2.3.1 Duty Location Assignment History

The PCT TRACON ATM entered on duty with the FAA on December 4, 1987, and began with initial training at the FAA Academy in Oklahoma City, OK. Below is a list of the PCT TRACON ATM's duty assignments in order according to their report date at each.

FAA Academy	December 4, 1987	Student
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ZNY ARTCC	March 1, 1988	Developmental, CPC, OS-IT, OS
PCT TRACON	November 10, 2009	NY/NJ/PHL Project Manager (detail) OAPM Manager (detail) OM, Airspace and Procedures Manager, ATM

2.4 PCT TRACON Mount Vernon (MV) Operations Manager (OM)

The PCT TRACON MV OM had only been working as OM since March 2024, but had been serving in some capacity at PCT TRACON since June 2010. The MV OM had extensive knowledge of operations pertaining to the operations between PCT TRACON and DCA ATCT.

2.4.1 Duty Location Assignment History

The PCT TRACON MV OM entered on duty with the FAA on June 4, 2010, and began with initial training at the FAA Academy in Oklahoma City, OK. Below is a list of the PCT TRACON MV OM's duty assignments in order according to their report date at each.

FAA Academy	June 16, 2010	Student
PCT TRACON	June 4, 2010	Developmental, CPC, TMC, STMC, Airspace/Procedures Manager, OM

2.5 Washington District (TEDC) Traffic Management Officer (TMO)

The TEDC TMO had not worked at DCA ATCT, however, had worked for many years in the DC area at other facilities including ZDC ARTCC, IAD ATCT, PCT TRACON, and the Air Traffic Control Command Center (ATSCC). They had extensive knowledge and involvement of the traffic management challenges and history of changes related to several facilities, specifically those involving PCT TRACON and DCA ATCT.

2.5.1 Duty Location Assignment History

The TEDC TMO entered on duty with the FAA sometime in 2007, the FAA was unable to provide definitive dates. The TEDC TMO reported to ZAU ARTCC before attending the FAA Academy in Oklahoma City, OK. Below is a list of the TEDC TMO's duty assignments in order according to their report date at each.

ZAU ARTCC	August 5, 2007	Developmental, CPC, TMC
FAA Academy	November 11, 2007	Student
ATSCC	October 21, 2012	NTMS
ZDC ARTCC	April 20, 2014	OS-IT, OS
PCT TRACON	November 13, 2016	STMC, TMO
IAD ATCT	February 3, 2019	OM
TEDC	March 15, 2020	TMO

3.0 DCA Helicopter Working Group (HWG) Personnel

3.1 Air Traffic Control Specialist - Detail

The ATC specialist from the FAA AJI Safety Integration Team had been on detail to FAA HQ from their permanent position as DCA ATCT OM since March 2023. They had previously served as a CPC, TMC, and OM at DCA ATCT and had extensive knowledge of the operations there and was a key member of the DCA HWG.

3.1.1 Duty Location Assignment History

The FAA ATCS entered on duty with the FAA on January 6, 2008, and worked at FAA HQ in an unrelated position until July 29, 2008, when they began initial ATC training at the FAA Academy in Oklahoma City, OK. Below is a list of the ATCSs duty ATC assignments in order according to their report date at each.

FAA Academy	July 29, 2008	Student
DCA ATCT	October 24, 2008	Developmental, CPC, TMC
BWI ATCT	December 9, 2018	OS-IT, OS
DCA ATCT	August 15, 2021	OM
FAA HQ	March 26, 2023	QA (detail)

3.2 Current DCA ATCT Operations Manager (OM)

The DCA ATCT OM was interviewed again, but this interview focused on the OM's role and involvement with the DCA HWG.

3.3 PCT TRACON Supervisory Traffic Management Coordinator (STMC)

The PCT TRACON STMC been working at PCT TRACON since September 2014, however prior to that had served as a CPC and OS at DCA ATCT. They had considerable knowledge of the involvement and happenings of the DCA HWG from previous years and had continued to be involved with the group since being at PCT TRACON.

3.3.1 Duty Location Assignment History

The PCT TRACON STMC entered on duty with the FAA on August 19, 2007, and was hired under Veterans Employment Opportunities Act of 1998 (VEOA) having previously served as an air traffic controller in the US Army from 2002-2007. Below is a list of the STMC's duty assignments in order according to their report date at each.

DCA ATCT	August 19, 2011	Developmental, CPC, OS-IT, OS
PCT TRACON	September 21, 2014	CPC-IT, CPC, TMC, STMC

4.0 Controller Workload

In post-accident interviews, the involved ATC personnel were asked to rate both the traffic volume and complexity for their position at the time of the event. On a scale from 1 to 5, with 1 being the lowest volume or least complexity, and 5 being the highest volume or greatest complexity, they provided the following:

The LC controller recalled the traffic volume being a 4 and thought that was a little heavier than normal. They recalled the traffic complexity to be about a 3 or 4, which they thought was normal, because of inbound aircraft with varying speeds, departures, helicopter traffic and gusty winds. They thought the complexity was decreasing just before the event because most of the departures were completed. They did not think they would have needed assistance from the supervisor unless one or two more helicopters would have checked in on frequency.

The ALC controller recalled the traffic volume as a 3 or 4 and the complexity as a 4. They thought that was normal for the time of day. They had not requested any relief during the time on position and did not recall any distractions around the time of the event.

The GC controller recalled the traffic volume as a 2-3 and the complexity as 1. They thought that was normal for that day and time and did not recall any distractions around the time of the event.

The CD controller recalled the traffic volume as a 2 and complexity as a 1. They thought it was normal for that time of day and did not recall any distractions around the time of the event.

The OS recalled the traffic volume as a 3 and the complexity as a 2. They thought it was normal for that time of day and did not recall any distractions around the time of the event. They had not requested a relief and noted that they were the only supervisor on duty.

5.0 Post-Accident Toxicology Testing

The LC and ALC controllers, and OS were administered a Department of Transportation (DOT) drug test on January 30, 2025, at 1508, 1658 and 1521, respectively. Results of all drugs tests were negative. Alcohol testing was not conducted for any of the controllers on position at the time of the accident. For full toxicology results and testing requirements, see the Medical Group Factual Report.

In response to the NTSB request for information about the process for determining whether to drug and alcohol test the controllers in the tower at the time of the event, the FAA responded via email on May 16, 2025, with the following:

Following the services rendered telcon (SRT), which occurred at 2330, the drug testing determination (DTD) occurred. The decision to drug test was made by the director of operations (DO) after any necessary consultation with the ATM, assistant general manager (AGM) and general manager (GM). At midnight on January 30, 2025, the ATM, after consulting with the district management and director's office, released the controllers from the facility. Regarding the decision to not alcohol test the controllers, the FAA stated *"No specific request for alcohol testing was made and it was uncertain if testing would be conducted that day or in the near future. Drug testing must be conducted within 5 days and Alcohol testing within 8 hours."* It was further stated that the guidance used to make the determination were the DOT Order 3910.1D and DOT Drug and Alcohol-Free Departmental Workplace Drug and Alcohol Testing Guide.⁵⁸

Table 4 below is a timeline of events related to the post-accident toxicology testing of the tested controllers.

Table 9. Post-accident toxicology testing timeline.

Event	Date	Time	Source
Accident	January 29, 2025	2048	NTSB
Services Rendered Telcon (SRT)	January 29, 2025	2330	FAA email ⁵⁹
Controllers released	January 30, 2025	0000	FAA email
Drug Testing Determination	January 30, 2025	0015	DOT Report ⁶⁰
DPC notification	January 30, 2025	0022	DOT Report
DPC notified 1-800-NUMBER	January 30, 2025	0049	DOT Report
Controllers notified	January 30, 2025	1330-1500	DOT Report
Controllers tested	January 30, 2025	1508-1658	Lab Report ⁶¹

H. FAA WILLIAM J. HUGHES TECHNICAL CENTER

The group formally requested a visit to the FAA William J. Hughes Technical Center (WJHTC) For Advanced Aerospace in Atlantic City, NJ. The WJHTC is the nation's premier federal aviation laboratory for advancing the National Airspace System (NAS) and sustaining its continued safe and efficient operations. The Center's industry partners, and technical workforce use state-of-the-art laboratories to enable research, engineering, development, test, and evaluation of advanced aviation technologies. These combined assets deliver capabilities that modernize and sustain the nation's airspace system while simultaneously making it the safest, most efficient aviation system in the world.

⁵⁸ A copy of this order is included in Attachment 27: DOT Order 3910.1D - DOT Drug and Alcohol Testing Guide.

⁵⁹ FAA email dated May 16, 2025.

⁶⁰ DOT Post-Accident/Incident Drug/Alcohol Test Collection Report dated January 30, 2025.

⁶¹ Abbott Federal Drug Testing Custody and Control Form for individuals tested, dated January 30, 2025.

Organizationally, the Technical Center Directorate (ANG-E) reports to the FAA's NextGen (ANG) staff office. The director of the Technical Center oversees and leads a campus comprised of approximately 3,000 civil servant and contract scientists, researchers, engineers and support staff who engage with stakeholders to research, design, develop, test, and implement innovative improvements to the aviation system.

1.0 Standard Terminal Automation Replacement (STARS) System Lab

In an effort to better understand the STARS operational and conflict alert parameters, system logic, and local adaptation at DCA ATCT, the group attended an interactive briefing with FAA STARS subject matter experts. After the briefing, the group went to the STARS laboratory and was provided with a playback of the DCA mid-air collision event. The WJHTC STARS lab provided the Continuous Data Recording (CDR) data from the event on their certified lab positions (STARS monitors). Subject matter experts from the FAA's terminal automation second level engineering office remained with the group afterward to answer questions from the group.

We discussed in depth the variance in the radar data that had been provided during the preliminary phase of the investigation. An important takeaway here, was the updated software tools that are currently used by the FAA for processing and analysis of radar data. Software that the STARS subject matter experts made very clear was instrumental in the accurate processing of data, and this is the software that the NTSB should be using to do this. This software was available for our use and the WJHTC managers would assist the NTSB with requesting access to these tools, as well as, providing basic training to the NTSB ATC team on their use.

2.0 Cockpit Simulation Facility (CSF) and Vertical Flight Aviation Safety Technologies (VFAST) Laboratory

The CSF/VFAST labs were able to put our group "into the cockpit" through simulation very near that of the accident helicopter. Group members were able to manipulate and control the helicopter through simulation in a full-scale helicopter flight deck simulator with the laboratory manager in the right seat. A short question answer session followed, and the team welcomed the opportunity to collaborate further with the FAA and NTSB teams in support of this investigation.

3.0 Airport Facilities Terminal Integration Laboratory (AFTIL)

The AFTIL team was able to provide event reconstruction through the current Adacel MaxSim⁶² tower simulator. This included reconstructions with simulated

⁶² Adacel Systems, Inc. is a current provider of tower simulators in use across the NAS by FAA, military and DoD, as well as FAA Collegiate Training Initiative (CTI) schools. DCA ATCT is equipped with the

viewpoints from the helicopter, the regional jet, and from the ATC tower. This reconstruction was extremely valuable, and provided insight into what may have been experienced by those involved in the event from a visibility standpoint. The AFTIL team indicated the ability to provide improved reconstruction work, or even development of “alternate scenarios” that may provide assistance in future safety enhancements or recommendations.

4.0 NAS Innovations and Emerging Concepts (NIEC) Laboratory

The NIEC houses the research and development of new and existing air traffic control technologies. The team showcased human factors technologies such as eye-tracking technology research into understanding how controllers react to air traffic in simulated radar and tower control environments. These technologies can be used to plug-in an almost endless array of scenarios into their proprietary mock tower, or radar labs. These could be used for accident/incident reconstruction, control equipment installation and design, advanced simulation training for implementation of new airspace or airport design layouts or understanding the impact of new procedures in ATC. The team was able to build this event into their mock tower for reconstructive playback similar to the AFTIL, but have also offered a more robust playback that could be built to not only review the event, but also to consider alternate scenarios changing altitudes, flight paths, etc. for the purpose of safety recommendation development. The group continues to work with this team in support of this investigation.

5.0 Surveillance Analysis Tools

The group was provided with a briefing on the various surveillance systems in use by ATC, with a heavy focus on ADS-B. It was learned during this briefing, that there is a program that would allow for direct access to the ADS-B surveillance data by trusted partners. There is a myriad of industry partners with access to these valuable data for research, testing, monitoring, education, etc. The value of direct access to these data for NTSB investigative work cannot be understated, and we are hopeful that there is a path forward to being a trusted government partner with the FAA in accessing this important data set.

I. DCA HELICOPTER WORKING GROUP

According to interviews and information provided by current and former members of the DCA HWG, origins of the group date back to before the COVID-19 pandemic at least. Though they did not have a formal mission statement or statement of work for the group, the general consensus from those interviewed was that the group was formed by DCA ATCT to address the challenges and safety concerns with

Adacel MaxSim tower simulator.

the high density of mixed helicopter operations within the DCA class B airspace. The group was an ATC body that would invite helicopter operators and other stakeholders to participate as necessary in reviewing and addressing identified challenges and concerns.

The DCA ATCT OM (still OS at the time) was the management lead for the DCA HWG since around 2022. The core group was made up of DCA ATCT personnel that included at least two NATCA controllers. Meetings were somewhat ad-hoc, and they would normally meet together just the few of them, identify challenges and concerns, then would invite the local helicopter operators to meet at the facility and discuss. This was an interactive exchange between ATC and helicopter operators to “bridge the gap between the cockpit and controllers.” The genesis of these meetings could be general concern about something that was trending, or even an event or events that had been reported via the mandatory occurrence reporting (MOR) process.

The last meeting prior to the accident had occurred on August 16, 2024; no notes or minutes were provided regarding what was discussed at this meeting. See the Helicopter Operations/Human Performance Group factual report for additional information provided during interviews with the helicopter operators.⁶³

1.0 Government Accountability Office Study - Aircraft Noise

In 2019 the Government Accountability Office (GAO) was asked by Congress to conduct a study on Aircraft Noise in the DC area. This study was conducted and a final report with recommendations published in January 2021.⁶⁴ According to the report highlights, GAO was asked to review issues related to helicopter flights and noise within the DC area and examine (1) what was known about helicopters and noise from flights in the DC area, and (2) the extent to which FAA and helicopter operators had taken action to address helicopter noise in the D.C. area. GAO reviewed statutes, regulations, policies, and documents related to helicopter noise. In conducting this study, they analyzed helicopter operations data from 2017-2019 and interviewed FAA officials, and representatives from 18 DC area helicopter operators.

2.0 Helicopter Symposiums

The regional administrator’s office organized and held at least three helicopter symposiums as a result of the GAO study mentioned above. The first symposium was held in April 2022; follow-up symposiums were held in June and November 2022. Participants included departments from within the FAA, such as the quality control

⁶³ All documents regarding the DCA HWG activities provided by the FAA are included in Attachment 39: DCA Helicopter Working Group Documents.

⁶⁴ Final report, GAO Study GAO-21-200, Aircraft Noise, published January 2021 is included in Attachment 28: GAO Study - Aircraft Noise - 2021

group, and operators conducting operation in and around DCA airspace. The purpose of the symposiums varied but included understanding operator requirements, educating operators on ATC procedures, and discussing traffic “hot spots” around DCA.

3.0 Proposed Operational Changes

According to those interviewed from the DCA HWG, the findings presented in the GAO study resulted in a series of helicopter symposia over the following, all giving energy to the efforts of the DCA HWG. There were at least two significant proposals mentioned or discussed by all those interviewed regarding the DCA HWG. The first being the proposed establishment of “hot spots” in the DCA class B, and the other being the proposed elimination of helicopter route 4 between Hains Point and the Wilson Bridge and/or establishment of an alternate helicopter routing that would allow for north/south transits through the DCA class B airspace. At the time of this writing, though spoken about by all interviewed, we were unable to find physical evidence of these proposals being denied. The information below was the information we were able to collect from the DCA HWG members.

3.1 Proposal - Establishment of Hot Spots

According to interviews conducted with members of the DCA HWG, all referenced the proposal by the group to establish airborne hotspots in aeronautical charting. The initial proposal was submitted in November 2022 and was also included in the Safety Risk Management Panel (SRMP) conducted in December 2022.⁶⁵ Specifically, the group proposed creating three hot spots (which are illustrated in figure 20 as well):

Hot Spot 1 (HS1): JPN/Memorial Bridge Transition

- “Pilots use caution for RWY 33 departures when DCA is in a north configuration; RWY 15/19 arrivals when in a south configuration.”
- “Use caution for JPN rotary traffic and route/zone transition area.”

Hot Spot 2 (HS2): Hains Point/South Capitol Region

- “Pilots use caution for rotary aircraft operating in/out and local pattern of Joint Base Anacostia-Bolling, South Capitol Street Heliport, US Park Police Eagle’s Nest.”
- “RWY 4 departures when DCA is in a north configuration.”
- “Multiple route transitions

Hot Spot 3 (HS3): Joint Base Anacostia-Bolling/Route 4/Power Plant Region

⁶⁵ FAA SRMP Workbook completed after conclusion of the December 2022 SRMP is included in Attachment 29: DCA ATCT SRMP Workbook – FEB-MAR 2023.

- “Pilots use caution for fixed wing arrivals to RWY 1 and to RWY 33 when DCA is in a north configuration; RWY 15/19 departures in a south configuration.”

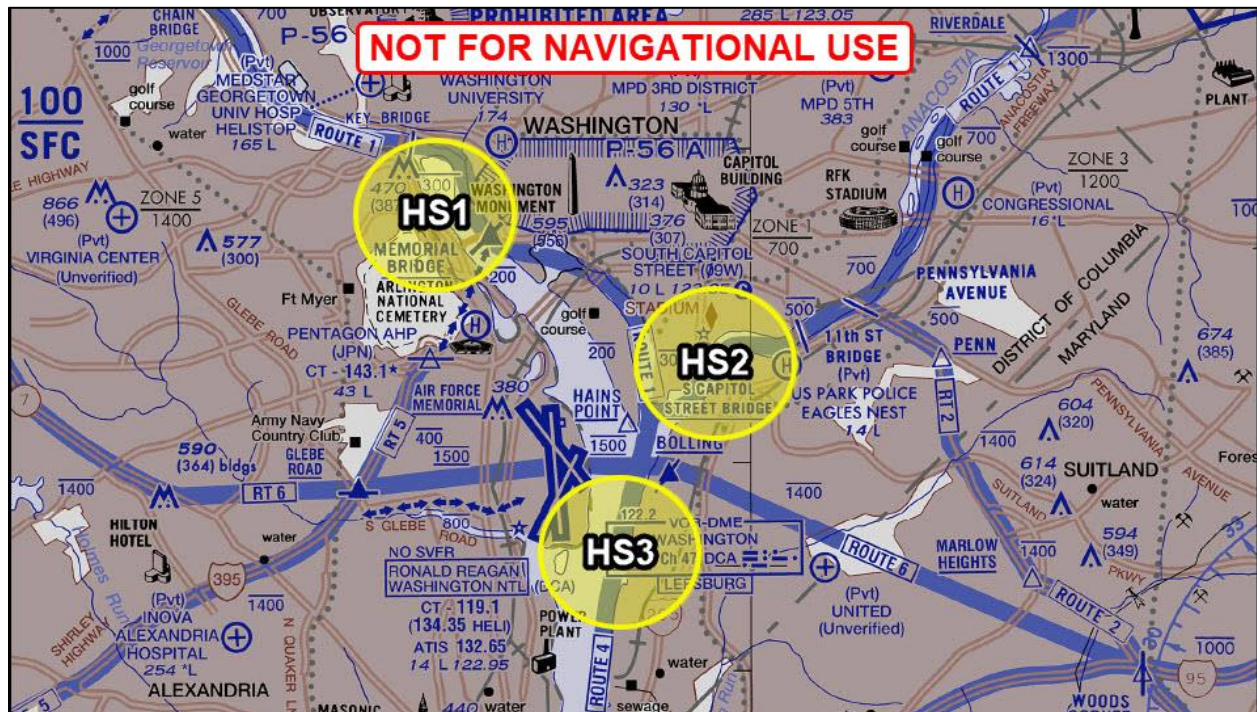


Figure 19. General areas of the DCA HWG's proposed hot spots 1-3.

According to information provided by those interviewed, the proposal was forwarded via the ATM and TEDC General Manager (GM). From the feedback they had received verbally from the ATM and/or GM, the proposal had been denied after being reviewed by the FAA's AJV office (where aeronautical charting is conducted). Status of this proposal could not be verified at the time of this writing.

3.2 Proposal - Helicopter Route 4 Change

According to interviews conducted with members of the DCA HWG, all referenced the proposal by the group to remove route 4 from around Hains Point to the Wilson Bridge, and establishing an alternate route for north/south transitions that would deconflict from the runway 1 and 33 approach corridor, as well as the runway 15 and 19 departure corridor.

According to information provided by those interviewed, the proposal was forwarded via the ATM and TEDC GM. According to their recollection, this proposal stopped at the GM's office who they specifically recalled stating something to the effect that it was "too political" and that they were not going to forward it further. Status of this proposal could not be verified at the time of this writing.

J. AIR TRAFFIC FLOW MANAGEMENT

1.0 Airport Arrival Rate (AAR) Information

AARs⁶⁶ are a significant factor in determining the complexity of airport and ATC operations. FAA Order JO 7210.3DD series, Facility Operations and Administration⁶⁷, outlined the responsibilities of those involved in developing or approving these rates, procedures for calculating the rates, and requirements for periodic review.

According to FAA Order JO 7210.3DD, the Deputy Director of System Operations (DDSO) is responsible to review and validate the airport primary runway configuration and associated AARs in February of each year, or at more frequent intervals if required. At the time of this writing, the ATC/HP group had not received any documentation showing evidence of these reviews.

1.1 Establishment of Operational Airport Arrival Rate (AAR)

In response to the ATC/HP group's request to the FAA for all documentation related to the most recent DCA Operational AAR computation, establishment, and/or review, the only items provided by the time of this writing were an FAA report titled Airport Capacity Profiles dated July 2014, with an associated information slides, and three reports regarding runway occupancy time (ROT) from different time periods between January 2015 and January 2016⁶⁸.

At the time of this writing, the ATC/HP group continued attempting to obtain more current data and/or documentation related to the most recent establishment, and annual reviews thereafter, of the DCA AAR.

1.2 Requested Change to DCA Airport Arrival Rate (AAR)

According to an FAA internal memorandum dated May 9, 2023⁶⁹, and forwarded on May 26, 2023, to the Washington district general manager (GM); the PCT TRACON ATM formally requested (in accordance with FAA Order JO 7210.3 series, "Facility Operation and Administration") a decrease to the existing AAR as follows:

⁶⁶ FAA Order JO 7210.3DD, defined AAR as: A dynamic parameter specifying the number of arrival aircraft that an airport, in conjunction with terminal airspace, can accept under specific conditions throughout any consecutive sixty (60) minute period

⁶⁷ A copy of this order is included in Attachment 30: FAA Order JO 7210.3DD - Facility Operations and Administration

⁶⁸ Copies of these reports are included in Attachment 31: DCA Airport Arrival Rate Information

⁶⁹ FAA Internal Memorandum included in Attachment 31: DCA Airport Arrival Rate Information

EXISTING AAR

Runway 19-32 32 (VMC) 30 (LVMC) 28 (IMC) 26 (LIMC)

Runway 01-36 36 (VMC) 34 (LVMC) 32 (IMC) 30 (LIMC)

REQUESTED AAR

Runway 19-32 28 (VMC) 26 (LVMC) 24 (IMC) 24 (LIMC)

Runway 01-36 32 (VMC) 30 (LVMC) 28 (IMC) 28 (LIMC)

The memo provided detailed justification for the requested change, including but not limited to:

- Fleet mix⁷⁰ over the previous decade
- Flight schedule increases that did not allow for use of reduced separation⁷¹
- Airspace and weather constraints
- Inability to meter⁷² traffic at DCA (see section 4.0 below)

According to records provided by the FAA, the GM responded to the PCT TRACON ATM on May 30, 2023, that they would need some time to discuss with the DDSO and TMO. No further email correspondence was located, however the PCT TRACON ATM stated during post-accident interviews that they were verbally informed by the GM that they would not be forwarding the request stating, "it's too political." The ATC/HP group has so far been unable to determine if the request was forwarded beyond the GM and continued to investigate this matter at the time of this writing.

1.3 Systemic Issue Review Report - January 30, 2025 (AAR)

A systemic issue review report (SYSIR) was initiated on January 30, 2025, immediately following the accident⁷³. The reason (details) for the SYSIR was "MIT is routinely not adhered to by PCT. DCA notes that PCT should consistently deliver 4 miles between arrivals." The SYSIR concluded and was checked "Concur & Elevate" and included information regarding AAR as well.

With regard to AAR, the SYSIR review team concluded that there was a systemic issue with the DCA AAR calculations for runways 01/19, resulting in over delivery of aircraft and subsequent non-compliance with associated MIT

⁷⁰ Fleet mix refers to the different types of aircraft servicing an airport, in this case DCA.

⁷¹ Reduced separation on final is outlined in FAA Order JO 7210.3DD, see Attachment 30: FAA Order JO 7210.3DD, Facility Operations and Administration.

⁷² Metering refers to a method of time-regulating traffic flows in the en route or terminal environments.

⁷³ A draft copy of SYSIR DCA-SI-2025/01/30-0002 is located in Attachment 32: MIT SYSIR - January 30, 2025. The associated .csv file containing numbers used during the SYSIR is included in Attachment 33: MIT SYSIR - January 30, 2025 - Data.

requirements. The SYSIR recommended the DCA AAR be adjusted to reflect compacted demand vs airspace constraints, fleet mix, and runway availability.

The SYSIR also cited arrival rates at the following airports utilizing similar runway configurations for comparison:

- San Diego International Airport (SAN) AAR - 24
- Palm Beach International Airport (PBI) AAR - 26/28
- Long Beach International Airport (LGB) AAR - 24

At the time of this writing, the ATC/HP group had received no updated information as to the status of the SYSIR elevation and continued to investigate.

1.4 Miscellaneous DCA Airport Arrival Rate Information

In June 2023, an MWA newsletter "On Good Authority"⁷⁴ reported on the DCA Airport's authority asking congress to defeat efforts that would add up to 28 new flights at DCA, increasing already heavy congestion on the "Busiest Runway in America." (see excerpt below). The group has added the ATC facility level for each of the ATCTs serving these airports for informational purposes.

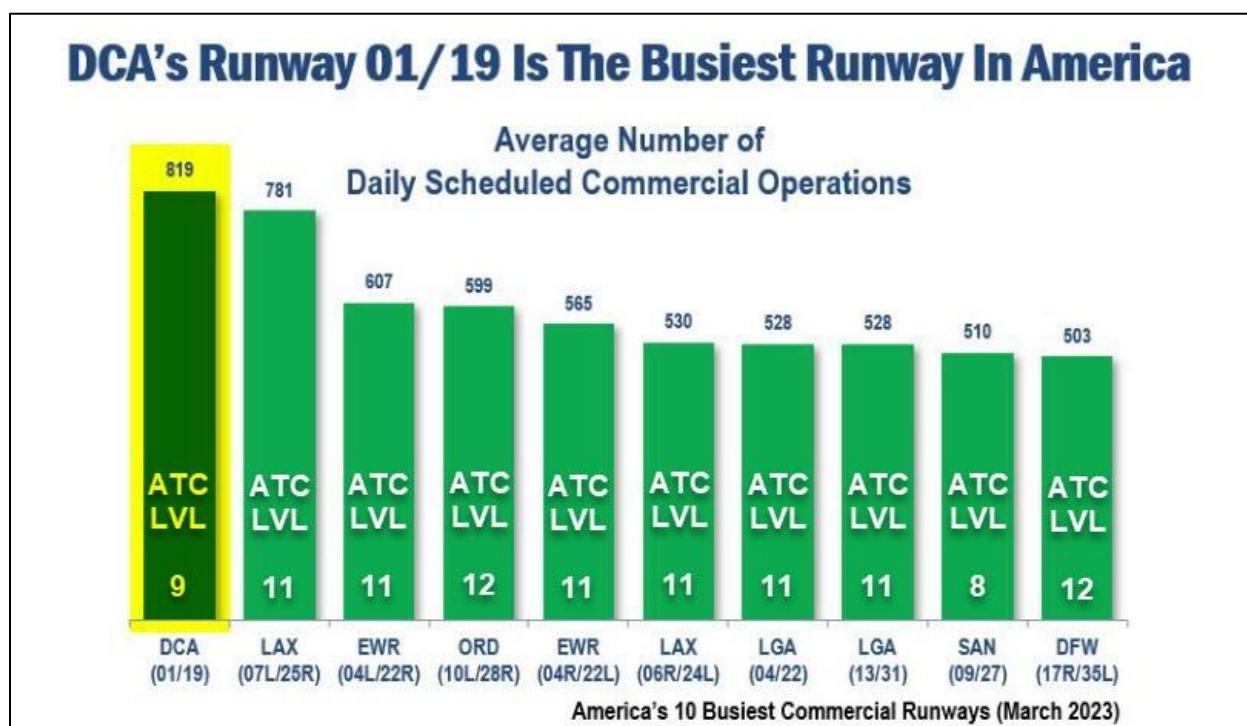


Figure 20. Graph illustrating the average number of daily scheduled commercial aircraft operations at the top 10 busiest runways in the United States. (Source: Published Airline Schedules for March 2023 via Cirium Diio mi, May 30, 2023, & historical public runway utilization data.).

⁷⁴ On Good Authority was a web-based newsletter published and distributed by the DCA MWA and discussed local events and news that affected the DCA MWA organization and local industry stakeholders.

2.0 Miles In Trail (MIT) Spacing

According to all DCA ATCT personnel interviewed, MIT spacing received from PCT TRACON had been a long-standing problem. DCA ATCT's airport surface area and airspace were both complex configurations with very little flexibility for adjustment. Notably, a phrase we heard repeatedly from DCA ATCT personnel during post-accident interviews and discussion was "we just make it work."

2.1 Letter of Agreement with PCT TRACON

Those interviewed also similarly explained that when in a north configuration (landing runway 1), especially during heavier periods of traffic, getting less than 4 miles separation to the runway threshold increased the workload at DCA ATCT significantly. Typically, letters of agreement (LOAs) between ATCTs and the radar facilities that feed their traffic flow, will prescribe how arrivals are to be handed off to the tower. It is common for these LOAs to prescribe a MIT requirement at a fix, boundary, or even to the runway threshold. The LOA between DCA ATCT and PCT TRACON was not explicit in this regard, and stated in part⁷⁵:

TOWER must:

- *Coordinate with TRACON (TMU) or MTV OS for final spacing separation required between successive arrivals for critical area protection, runway conditions, weather, and other factors.*
- *Coordinate with TRACON (TMU) for changes in runway acceptance rates.*

TRACON must:

- *Comply with final spacing requirements as coordinated between the Tower OS and the TRACON MTV/OS/TMU.*

According to interviews conducted with DCA ATCT and PCT TRACON personnel, the agreed upon mitigation between the facilities was that the DCA ATCT was required to call and implement arrival restrictions of 4 MIT via TMU. Both DCA ATCT and PCT TRACON personnel stated that this meant 4 miles to the runway threshold and not to an arrival fix or airspace boundary. It was also stated that even during times of entered restrictions, DCA ATCT would routinely continue to receive aircraft with less than 4 MIT. Both also admitted that there had been a shared frustration for both facilities over the years regarding MIT issues. The primary challenge for DCA ATCT was having room to get departures out without backing up due to a complex and limited surface area. The challenge for PCT TRACON was having adequate airspace to hold aircraft inbound to DCA ATCT (and the other

⁷⁵ A copy of the DCA ATCT / PCT TRACON LOA is included in Attachment 7: DCA ATCT Letters of Agreement.

airports they service approach control services to) due to complex and limited airspace.

2.2 Traffic Flow Mitigations

The challenges of efficient traffic flow and ability to meet capacity demand required continual mitigations on the part of tower personnel at DCA ATCT. This included routine use of conditional departure instructions, as well as offloading⁷⁶ arrival traffic to another runway to build in spacing.

2.2.1 Conditional Departure Instructions

According to interviews with DCA ATCT personnel, as well as review of a number of ATC audio recordings from periods of heavier traffic load, conditional departure instructions were commonplace. A common example would be:

"Airplane 123, runway one, line up and wait no delay, keep your engines spooled up, traffic three mile final for runway three three."

"Airplane 123, runway one, traffic two mile final for runway three three, cleared for immediate takeoff."

There were a number of occasions where multiple departures were all given conditional departure instructions similar to those above, one after another.

2.2.2 Offloading Arrivals to Runway 33

According to interviews with DCA ATCT personnel, as well as review of other audio recordings and personal observation by the ATC/HP group, offloading arrivals to another runway was common to build space. Specifically, during periods of heavier traffic flow and in a north configuration (landing runway 1), offloading every few arrivals to runway 33 to build space for departures was necessary and was frequently done. This is what occurred on the evening of the accident.

This was the scenario on the night of the accident. There were multiple aircraft holding and taxiing for departure, and multiple aircraft on final to land runway 1. In order to keep ground traffic from backing up, the LC controller was soliciting flight crews willing to accept runway 33 instead of runway 1. In this case, the aircraft immediately preceding JIA5342 was asked, and the flight crew declined. Then, JIA5342 was asked, and they agreed to accept the runway change to runway 33. The crew was then instructed to change to and circle to runway 33.⁷⁷

⁷⁶ Offloading is a term often used to describe the practice of moving arrivals planned to one runway, to an alternate runway in order to build extra space allowing time to get departures out in between closely spaced arrivals.

⁷⁷ According to the FAA Airman's Information Manual (AIM) Pilot/Controller Glossary (included in Attachment 34: FAA Aeronautical Information Manual (AIM)) - Circle to runway (runway number) is

The change from runway 1 to runway 33 required flight crews to break off from the final approach course they were on and circle to the final approach course of the other runway. When performing a circling maneuver to an alternate runway, the pilot continues on the final approach course until they have the runway they are planning to circle to in sight, then given instructions to circle to the alternate runway. The flight crew will then circle toward the alternate runway's final approach course, and once aligned, continue descent to land normally.

2.2.3 Systemic Issue Review Report - January 30, 2025 (MIT)

As mentioned in section K, 1.3 above, a SYSIR was conducted on January 30, 2025, and addressing the systemic issue in delivering 4 MIT, with findings of "Concur & Elevate."

Regarding MIT, the SYSIR review team concluded that PCT TRACON routinely delivered 4 MIT to the transfer of control point (TCP), but not to the runway threshold. According to the SYSIR, data showed that there was an average loss of 1.017 miles between the TCP and runway threshold which resulted in a 4 MIT compliance rate of 63.13% for runway 1. This led to runway 33 having to be used to maintain throughput and avoid gridlock⁷⁸. The SYSIR concluded similar findings for runway 19.

The SYSIR recommended that a 5 MIT was necessary instead of the current 4 MIT to ensure they received the 4 MIT needed at runway threshold.

At the time of this writing, the ATC/HP group had received no updated information as to the status of the SYSIR elevation and continued to investigate.

3.0 Slot Program

The FAA uses runway slots to limit scheduled air traffic at certain capacity constrained airports. In the U.S., those airports are John F. Kennedy International Airport (JFK), LaGuardia Airport (LGA), and DCA. At DCA, slot controls that are equivalent to Level 3⁷⁹, are in place pursuant to 14 CFR part 93 subparts K and S to govern operations daily, from 0600-2359. During slot-controlled hours, regularly scheduled operations are limited to 48 per hour by regulation.

used by ATC to inform the pilot that he/she must circle to land because the runway in use is other than the runway aligned with the instrument approach procedure. When the direction of the circling maneuver in relation to the airport/runway is required, the controller will state the direction (eight cardinal compass points) and specify a left or right downwind or base leg as appropriate; e.g., "Cleared VOR Runway Three Six Approach circle to Runway Two Two," or "Circle northwest of the airport for a right downwind to Runway Two Two."

⁷⁸ Gridlock in ATC refers to a situation where the flow of air traffic is significantly hindered, restricted, or stopped altogether, leading to delays and congestion in the NAS.

⁷⁹ Level 3 is assigned where infrastructure is inadequate to meet demand and there is significant potential for delays requiring mandatory slot controls.

According to an FAA internal memo from the ATO to the Office of Government and Industry Affairs (AGI) dated May 25, 2023, the FAA received a request from Capital Access Alliance (CAA) to increase the slots at DCA "beyond-perimeter." The FAA ATO stated that as long as the flights were operated beyond-perimeter AND did not increase the overall number of flights, then FAA took non position on the flights being operated within or beyond-perimeter.

However, the ATO stated that the requesters recommendation of adding 20-25 slot pairs (40-50 new slots) and claiming to show that DCA could support the increase was "flawed," and stated that between January 2022 and April 2023:

- *DCA had the 5th largest number of Ground Delay Programs (GDP) and Ground Stops (GS) in the NAS.*
- *DCA had received 69% more GDPs and GSs than BWI and IAD combined.*
- *DCA ranked 10th among the most delayed airports in the NAS, but only 19th in number of operations.*
- *DCA experiences an average delay of 67 minutes for 20% of their departures and 22% of their arrivals.*
- *DCA was in the top 5 for GDP and GS counts among the nation's "Core 30" (the nation's 30 busiest airports)*
- *DCA had more one loop airborne holds (less than 15 minutes) than all other airports and ranked 3rd in reportable airborne holding events (greater than 15 minutes) per arrival.*

The memorandum went on and concluded with the FAA ATO clarifying a couple of the claims made in CCA's request:

CCA Claim:

ASV was increasing, so more slots were needed was not accurate.

FAA Clarification:

In fact, slot limits were based on airport capacity, not airport demand, and the FAA imposed slot limits when demand exceeded capacity.

CCA Claim:

DCA had a minute more padding than their peers showing that DCA outperformed.

FAA Clarification:

DCA having an extra minute of block time buffer compared to other major airports actually suggests underperformance, as airlines need the padding to ensure an 80% on time rate.

Additionally, in another FAA memo from the PCT ATCT to the Reagan National Community Noise Working Group on May 23, 2023, citing exacerbation of

current operational challenges and escalation in noise impacts; stated they did not support in an increase to the DCA Slot Program.⁸⁰

4.0 Time Based Flow Management

Time-based flow management (TBFM) is a foundational decision support tool (DST) for time-based management in the en route and terminal environments. TBFM's core function is the ability to schedule aircraft within a stream of traffic to reach a defined constraint point (e.g., meter fix/meter arc) at specified times, creating a time-ordered sequence of traffic. The scheduled times allow for merging of traffic flows, while minimizing coordination, reducing the need for vectoring/holding, and efficiently utilizing airport and airspace capacity. The TBFM schedule is based on current aircraft estimated time of arrival at key defined constraint points based on wind forecasts, aircraft flight plan, the desired separation at the constraint point and other parameters. The schedule applies time-based spacing only when needed to maintain the desired interval at one or more constraint points.

According to several of those interviewed, TBFM had been installed for years that would allow for metering to DCA, and controllers had been trained – however it had not been activated. According to records provided by the FAA⁸¹ the most recent status request was made by ZDC ARTCC in February 2023. The first entry and last two most recent entries are provided below:

February 23, 2023 – Initial request

ZDC requests Ops Team support with reviewing/updating the current TBFM arrival system for DCA. Ideally to be used for scheduling purposes at least within ZDC and the surrounding 1st tier facilities (ZJX/ZTL/ZID/ZOB/ZNY/ZBW).

January 25, 2025 – Comment 4

Per the comments on this ticket, I have received agreement from ZJX/ZTL/ZID/ZOB/ZNY/ZBW as well as PCT though they were not listed to participate in future scheduling to DCA, so that this ticket may proceed to the next steps. Upon request, I can provide the email receipts as I am unable to attach here.

February 10, 2025 – Comment 5

In response to site update, this item was revisited at the Gov Board meeting on 2/6/2025 with AJT, NATCA, Program Office, and SLE present. Disposition: Due to budget constraints and other program priorities, this effort is currently on hold until further notice.

⁸⁰ A copy both memos are included in Attachment 35: FAA Memos – Expanding Slots at DCA.

⁸¹ Copy of FAA log documenting status of request by ZDC ARTCC is included in Attachment 36: Time Based Flow Management (TBFM) Information.

The ATC/HP group continues to investigate the details surrounding this issue area and are awaiting further historical documentation.

K. RELATED RECOMMENDATIONS

On November 15, 2016, the NTSB issued two safety recommendations to the FAA following two midair collisions. The first accident occurred on July 7, 2015, about 1101 eastern daylight time, when a Cessna 150M and a Lockheed Martin F-16CM, operated by the US Air Force, collided in midair near Moncks Corner, South Carolina. The private pilot and passenger aboard the Cessna died. The second accident occurred on August 16, 2015, about 1103 Pacific daylight time, when a Cessna 172M and an experimental North American Rockwell NA265-60SC Sabreliner collided in midair about 1 mile northeast of Brown Field Municipal Airport (SDM), San Diego, California. The pilot (and sole occupant) of N1285U and the two pilots and two mission specialists aboard the Sabreliner died.

Safety Recommendation A-16-051: Include the July 7, 2015, and August 16, 2015, midair collisions as examples in your instructor-led initial and recurrent training for air traffic controllers on controller judgment, vigilance, and/or safety awareness.

Safety Recommendation A-16-052: Brief all air traffic controllers and their supervisors on the air traffic control errors in the July 7, 2015, and August 16, 2015, midair collisions.

On May 24, 2023, the NTSB classified these recommendations Closed-Unacceptable Action.

L. AIR TRAFFIC CONTROL PROCEDURAL REFERENCES

1.0 General Control

FAA Order JO 7110.65AA, Paragraph 2-1-1
FAA Order JO 7110.65AA, Paragraph 2-1-2
FAA Order JO 7110.65AA, Paragraph 2-1-4
FAA Order JO 7110.65AA, Paragraph 2-1-5
FAA Order JO 7110.65AA, Paragraph 3-1-1
FAA Order JO 7110.65AA, Paragraph 3-1-9
FAA Order JO 7110.65AA, Paragraph 3-1-10
FAA Order JO 7110.65AA, Paragraph 5-1-22
FAA Order JO 7110.65AA, Paragraph 5-2-24

2.0 Resource Management

FAA Order JO 7210.3DD, Paragraph 2-6-5
FAA Order JO 7210.3DD, Paragraph 10-3-8
DCA ATCT Order 7110.2L, Paragraph 11-1
DCA ATCT Order 7110.2L, Paragraph 11-20

3.0 Class B Service Area / SFRA

FAA Order JO 7110.65AA, Chapter 7, Section 9
FAA Order JO 7210.3DD, Paragraph 12-1-1
FAA Order JO 7210.3DD, Paragraph 12-1-5
FAA Order JO 7110.65AA, Paragraph 9-2-10
FAA Order JO 7110.65AA, Paragraph 9-2-12

4.0 Separation and Visual Separation

FAA Order JO 7110.65AA, Paragraph 7-2-1
FAA Order JO 7110.65AA, Paragraph 7-9-4
FAA Order JO 7110.65AA, Paragraph 7-9-5
FAA Order JO 7210.3DD, Paragraph 10-3-9
FAA AIM, Paragraph 4-4-14
FAA AIM, Paragraph 4-4-15
DCA ATCT Order 7110.2L, Paragraph 4-10
DCA ATCT Order 7110.2L, Appendix B

5.0 Traffic Advisories and Safety Alerts

FAA Order JO 7110.65AA, Paragraph 2-1-6
FAA Order JO 7110.65AA, Paragraph 2-1-21
FAA Order JO 7110.65AA, Paragraph 3-1-6
FAA Order JO 7110.65AA, Paragraph 5-1-4
FAA Order JO 7110.65AA, Paragraph 5-14-6
FAA Order JO 7110.65AA, Paragraph 7-9-5
FAA Order JO 7210.3DD, Paragraph 10-5-3
DCA ATCT Order 7110.2L, Paragraph 4-10
DCA ATCT Order 7110.2L, Paragraph 9-2

6.0 Tower Team Responsibilities

FAA Order JO 7110.65AA, Paragraph 2-10-3
FAA Order JO 7210.3DD, Paragraph 10-1-1
DCA ATCT Order 7110.2L, Paragraph 4-13
DCA ATCT Order 7110.2L, Paragraph 10-2
DCA ATCT Order 7110.2L, Paragraph 11-1

M. LIST OF ATTACHMENTS

Attachment 1: ATC Audio Recordings
Attachment 2: FAA Aircraft Accident Package
Attachment 3: DCA Approach Procedures
Attachment 4: ADS-B Data
Attachment 5: STARS Radar Data
Attachment 6: PDARS Studies
Attachment 7: DCA Letters of Agreement
Attachment 8: Memorandum of Record - AJV-A2
Attachment 9: DCA ATCT Workstation Documentation
Attachment 10: DCA ATCT Staffing Documentation
Attachment 11: DCA ATCT SOP - Multiple Versions and Notices
Attachment 12: DCA ATCT TOP Data
Attachment 13: Individual Training and Proficiency Records
Attachment 14: FAA Order JO 3120.4S - Air Traffic Technical Training
Attachment 15: DCA ATCT 3120.4J - DCA ATCT Facility Training
Attachment 16: Air Traffic National Training Initiative - January 2025
Attachment 17: DCA ATCT 2024 Refresher Training Plan
Attachment 18: DCA ATCT Pertinent Training Information
Attachment 19: FAA Order 1100.161B - Air Traffic Safety Oversight
Attachment 20: FAA Order JO 1000.37C - ATO Safety Management System
Attachment 21: FAA Order JO 7200.20B - Voluntary Safety Reporting Program
Attachment 22: FAA Order 8040.4C - Safety Risk Management Policy
Attachment 23: FAA ATO Safety Management System Manual
Attachment 24: Interview Transcripts
Attachment 25: Involved Controllers - Hours Worked
Attachment 26: DCA ATCT Facility Logs and Records
Attachment 27: DOT Order 3910.1D - DOT Drug and Alcohol Testing Guide
Attachment 28: GAO Study - Aircraft Noise - 2021
Attachment 29: DCA ATCT SRMP Workbook - Feb-Mar 2023
Attachment 30: FAA Order JO 7210.3DD - Facility Operations and Administration
Attachment 31: DCA Airport Arrival Rate Information
Attachment 32: MIT SYSIR - January 30, 2025
Attachment 33: MIT SYSIR - January 30, 2025 - Data
Attachment 34: FAA Aeronautical Information Manual (AIM)
Attachment 35: FAA Memo - Expanding Slots at DCA
Attachment 36: Time Based Flow Management (TBFM) Information
Attachment 37: FAA Order JO 7110.65AA - Air Traffic Control
Attachment 38: FAA Order JO 7210.632A - ATO Occurrence Reporting
Attachment 39: DCA Helicopter Working Group Documents

Submitted by:

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